Natural Origin IV — Atoms, Elements, Electrons, Electromagnetic Waves, and Photons

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Abstract

This paper proposes and systematically establishes a unified theoretical framework in which helical space flow is taken as the sole physical ontology, providing an integrated reconstruction of atomic structure, electromagnetism, quantum entanglement, communication, and computation. The theory starts from a single axiom: space, as a continuous geometric entity, is intrinsically in motion, and the magnitude of its geometric velocity field is identically equal to the speed of light $|\mathbf{g}| = c$. On this basis, a three-state structure—enclosure, propagation, and re-enclosure—together with single-channel and principal-rotation stability constraints, is introduced, forming a geometric physical system that requires neither quantum-mechanical operators, postulates, nor independent field ontologies.

At the microscopic level, the paper demonstrates that electrons, protons, and neutrons are all single-channel enclosure states organized around a unique principal rotation direction. From this result, the atomic shell-capacity theorem $N(n) = 2n^2$ is rigorously derived, reducing the periodic table of elements to a geometric dimensional invariant of the global single-channel standing-wave space, rather than an empirical rule or a postulated set of quantum numbers.

At the propagation level, electromagnetism is reinterpreted as a geometric projection of helical space flow in the continuous limit. Maxwell's equations are shown not to be fundamental physical axioms, but effective theorems valid within the propagation regime. Their domain of applicability and breakdown boundaries are explicitly identified, and quantization is clarified as a necessary manifestation at re-enclosure endpoints, rather than a corrective supplement to classical theory.

At the level of correlation and information, quantum entanglement is formalized as the non-decomposability of a global double-helical enclosure shell. It is shown that the correlation function $-\cos\theta$ arises uniquely from continuous geometric projection under endpoint discretization, and that the apparent "violation" of Bell inequalities originates from the failure of their separability assumption within this geometric structure, rather than from nonlocal superluminal causation.

At the engineering and computational level, communication and computation are reformulated from the traditional bit–probability paradigm into problems of phase writing, propagation, re-enclosure discrimination, and continuous phase-structure evolution. Physical feasibility boundaries for field-based communication and continuous computation are derived, clarifying which correlations can be established and which forms of control are fundamentally impossible.

In summary, this work completes a full geometric closed loop from the enclosure axiom of space → atomic structure → electromagnetic propagation → quantum entanglement → communication and computation, offering a unified physical description that requires neither independent field ontologies, quantum postulates, nor statistical interpretations, and providing a new theoretical pathway for understanding the physical foundations of matter, information, and computation.

Introduction — Geometric Ontology, Principal-Axis Convention, and the Cross-Scale Enclosure Axis Theorem

Before entering the main derivations, it is necessary to give a consolidated explanation of the geometric ontology, spatial orientation conventions, and core structural principles employed throughout this work. This introduction introduces no new physical assumptions; rather, it brings forward, unifies, and anchors geometric facts that are repeatedly used later but distributed across different chapters, in order to avoid misunderstandings at the level of orientation or ontology across scales and stages of argument.

1. Rotation is not a Property but an Ontology: A Common Geometric Prototype of Electrons, Protons, and Neutrons

Within the framework of this paper, electrons, protons, and neutrons are not "point particles carrying a spin attribute," but rotational enclosure structures formed by space itself under stable conditions.

Here, "rotation" refers neither to classical rigid-body motion nor to a quantum-state label, but to the geometric organization of helical space flow.

Their unified ontological descriptions are as follows:

- Electron: a minimal-scale single-channel enclosure state, whose interior consists of a tightly closed double-helical structure;
- Proton: a higher-order single-channel enclosure state with a more complex internal enclosure hierarchy, still organized around a unique principal rotation direction;
- Neutron: a stable state formed by introducing a phase-sign compensation shell on top of the proton's single-channel enclosure structure, while preserving the same principal rotation direction.

Thus, the differences among electrons, protons, and neutrons do not lie in the "presence or absence of spin," but in their enclosure hierarchy and phase structure. Geometrically, all three share the same fundamental prototype.

2. Spatial Orientation of the Electron Model: Double-Helical Closed Ring, Flux Tube, and Principal Axis

To eliminate all spatial ambiguities concerning flux tubes, propagation axes, measurement axes, and principal rotation directions, this paper establishes a global and unambiguous convention for the minimal enclosure model of the electron. The same convention applies to protons and neutrons, differing only in enclosure hierarchy.

(1) Plane of the Double-Helical Closed Ring

The minimal stable enclosure structure of an electron is a double-helical closed ring:

- The closed ring lies entirely within a two-dimensional plane;
- This plane is uniformly defined as the XY plane;
- The geometric center of the closed ring coincides with the coordinate origin.

Hence, the electron's rotational enclosure is represented as a double-helical closed ring lying in the XY plane.

(2) Rotation Direction and the Right-Hand Rule

The double-helical closed ring undergoes stable rotation along the ring direction. A single,

unambiguous right-hand rule is adopted to define the rotation sense and magnetic polarity:
If one grips the rotating double-helical ring with the right hand such that the fingers follow the rotation direction, the direction indicated by the thumb defines the flux-tube emission direction (North pole).
(3) Flux-Tube Principal Axis and the Z Axis
The emission direction uniquely determined by the right-hand rule defines the principal axis of the magnetic flux tube. In the global coordinate convention of this work, this direction is defined as the Z axis.
Therefore:

- The flux tube is emitted along the +Z direction (North pole);
- The −Z direction corresponds to an absorption port (South pole);
- The flux-tube axis is perpendicular to the XY plane of the double-helical ring;
- The axis passes through the geometric center of the enclosure structure.

This $XY - Z$ orientation is the only spatial convention adopted throughout the entire work.

3. Physical Meaning of the Single Channel: The Minimum-Cost Direction of Stable Enclosure

The flux tube formed along the principal rotation direction is not an auxiliary feature, but the geometric extension of the rotational enclosure structure into external space. This extension is termed the single channel.
The single channel is not an arbitrarily chosen axis; it is the minimum-cost direction of the enclosure geometry in the sense of an energy functional. Attempting to introduce a second independent principal axis or parallel channel within the same structure inevitably leads to:

- discontinuities in rotation sense;
- multivalued phases;
- or geometric conflicts at channel junctions.

The result is enclosure instability and decay into a propagation state. Therefore:
Stable enclosure ⇔ Unique principal rotation ⇔ Indivisible single channel.

4. Hierarchical Inheritance of the Principal Axis: From Microscopic Rigidity to Macroscopic Equivalence

At the microscopic level (electrons, protons, neutrons), the principal rotation axis is a rigid, unique, and well-defined geometric object. At higher scales, its manifestation changes hierarchically:

- Nuclear level: principal axes of individual nucleons undergo angular averaging under strong n and Ω contributions, forming an effective nuclear principal axis;
- Atomic level: the electron cloud does not compete for the principal axis, but forms phase-readout configurations around the nuclear axis;
- Molecular and material level: atomic principal axes become locked, compromised, or statistically aligned through bonding and lattice structures, appearing as bond directions, crystal axes, easy-magnetization axes, and anisotropy directions;
- Astronomical level: the principal axis emerges statistically or macroscopically as rotation axes, magnetic axes, and jet directions.

5. Cross-Scale Enclosure Axis Theorem

Based on the unified geometric facts above, the following system-level theorem is established as the structural backbone of this work:

Theorem (Central Axis Theorem of Enclosure)

From particles to celestial bodies, any enclosure structure that can exist stably over long timescales and interact with its environment must possess a unique effective principal axis. This axis passes through the geometric center of the enclosure and is uniquely selected by the principle of minimum enclosure cost.

At microscopic scales, this axis appears as a rigid geometric rotation axis; at macroscopic scales, it is inherited and manifested statistically, structurally, or effectively. Phase readout, energy transport, and structural reconfiguration can occur only along this axis or its equivalent directions.

Clarification against Misinterpretation

The "principal axis" referred to here is neither a classical rigid-body rotation axis nor a simple superposition of quantum spin vectors, but the minimum-cost direction of enclosure geometry under constraint.

6. Role of This Introduction in the Full Text

The purpose of this introduction is to:

- provide a unified geometric-orientation anchor for Chapter 6 (atomic structure and shell capacity), Chapter 8 (electromagnetic propagation), Chapter 10 (entanglement), and Chapters 11–12 (communication and computation);
- prevent readers from misinterpreting the Z axis, propagation axis, or measurement axis as arbitrary coordinate choices;
- clarify that the paper does not introduce different models chapter by chapter, but develops a single principal-axis principle across multiple scales.

0.2 Three Fundamental Structural Parameters $\boldsymbol{\phi}, \boldsymbol{\Omega},$ and $\boldsymbol{n}$: Definitions, Dimensions, and Division of Labor in the XYZ Geometric Framework

In the following chapters, the phase ϕ, the rotational frequency Ω, and the enclosure density n will be repeatedly used as the basic parameters for describing physical states. To avoid misinterpreting them as variables of conventional field theory or as quantum-mechanical operators, it is necessary here to give a consolidated account of their geometric definitions, spatial orientations, dimensions/units, and ontological roles. This section introduces no new physical assumptions; it only unifies and anchors the geometric variable system already adopted throughout the paper.

0.2.1 The Sole Ontological Object and Coordinate Convention

The only ontological object in this paper is the geometric velocity field of helical space flow, $\mathbf{g}(\mathbf{r}, t), \quad |\mathbf{g}| = c$,where c is the speed of light.

To eliminate all spatial ambiguities, the paper adopts the following fixed geometric conventions throughout:

- XY plane: the rotation plane of the minimal stable enclosure (the plane containing the electron's double-helical closed ring);
- Z axis: the enclosure principal axis and the flux-tube axis, uniquely determined by the

right-hand rule;

- the coordinate origin is located at the geometric center of the enclosure structure.

All "propagation directions," "principal-axis directions," and "measurement-axis directions" must be understood relative to this XYZ geometric framework, rather than as arbitrary coordinate selections.

0.2.2 Phase ϕ: A Direction Marker in the XY Plane

Definition (Phase ϕ).

The phase ϕ describes the directional state of helical space flow at a given position and time, corresponding to the orientation of $\mathbf{g}$ on the unit sphere.

In the minimal enclosure and propagation states:

- variations of ϕ are manifested mainly as angular differences within the XY plane;
- i.e., ϕ specifies "which angle" the space flow points to within the rotation plane.

Physical roles:

- provides distinguishability and information bearing;
- determines interference, correlations, and readout differences.

Dimension and unit:

- dimensionless (an angular variable);
- unit: radian (rad).

In the propagation state, ϕ can evolve continuously; at re-enclosure endpoints, ϕ becomes thresholded and discretized.

0.2.3 Rotational Frequency Ω: Rotational Strength about the Z Axis

Definition (Rotational frequency Ω).

Ω describes the rotational intensity of helical space flow around the enclosure principal axis (the Z axis), i.e., the geometric rotation frequency required to maintain rotational continuity.

Within the unified model of this paper:

- the rotation axis of Ω is strictly along the Z direction;
- the electron's double-helical closed ring lies in the XY plane;
- grasping the ring with the right hand along its rotation direction, the thumb points to +Z (the flux-tube emission port, N pole).

Physical roles:

- determines the principal axis and flux-tube direction;
- manifests as magnetism, magnetic moment, and the propagation axis;
- after de-enclosure, the inherited direction of Ω becomes the Z axis of electromagnetic propagation.

Dimension and unit:

- inverse time;
- SI unit: s^{-1} (or $\mathrm{rad\,s}^{-1}$, treated equivalently in this paper).

0.2.4 Enclosure Density n: Closure along the Principal Axis and Manifestation as Matter

Definition (Enclosure density n).

n describes where, and with what strength, helical space flow is forced to close along the principal-axis direction, thereby manifesting as a stable object.

In the continuous limit, n is related to the divergence of $\boldsymbol{g}$ (see Chapter 8): $n \propto \nabla \cdot (\mu_g \mathbf{g})$
Physical roles:

- $n > 0$: enclosure occurs and an object appears;
- $n = 0$: the flow remains in a propagation state and no mass is manifested;
- the spatial gradient of n (mainly along Z) determines the direction of gravitational fall.

Dimension and unit:

- mass density;
- SI unit: $\mathrm{kg\,m^{-3}}$.

0.2.5 Orthogonal Division of Labor of the Three Parameters in the XYZ Framework

Within the fixed XYZ geometric framework, ϕ, Ω, and n have a clear spatial division of labor:

Parameter	Primary direction	Question addressed	Physical manifestation	Dimension
ϕ	XY plane	How to distinguish directions/states	Interference, entanglement, information	Dimensionless
Ω	Z axis	About which axis the flow rotates	Magnetism, propagation direction	$\mathrm{s^{-1}}$
n	Closure along Z	Where matter appears	Mass, gravitation	$\mathrm{kg\,m^{-3}}$

Together, they form a minimal complete description:

- ϕ controls distinguishability within the plane;
- Ω controls axial rotation and the channel;
- n controls stable enclosure along the axis.

0.2.6 Clarification to Avoid Misinterpretation (Coordinate-Related)

It is essential to strictly distinguish the following:

- the XY plane is not an arbitrary choice, but the rotation plane of the minimal enclosure;
- the Z axis is not merely a computational coordinate axis, but the enclosure principal axis and flux-tube axis;
- ϕ, Ω, and n are not conventional fields or quantum operators, but structural parameters of helical space flow.

Any reinterpretation of these variables as “free coordinate choices” or “quantum-state labels” is inconsistent with the ontological setting of this paper.

0.2.7 Role of This Section in the Full Text

The purpose of this section is to:

- clarify the geometric meanings of all symbols and directions before entering atomic structure (Chapter 6);
- provide a unified explanation for the Z axis used in electromagnetic propagation (Chapter 8) and for the global rotational variables in entanglement (Chapter 10);
- prevent readers from mistaking later appearances of axial directions, polarization, or measurement orientations as ad hoc conventions.

Chapter 1 Ontology, Axioms, and the Computational Template of the Enclosure Universe

1.1 Ontological Axiom: The Universe Consists Only of Objects and Space

This book adopts the following single ontological axiom: the universe contains only two kinds of entities—objects and space.

- Objects: stable enclosure structures formed by space under specific conditions;
- Space: a continuously existing geometric continuum that can propagate, rotate, and be enclosed.

There is no third ontology:

- no independent "field entity";
- no "pure energy entity";
- no "information ontology."

All "fields," "particles," "interactions," and "energies" are external manifestations of space under different states of enclosure, propagation, and re-enclosure.

1.2 Physical Definition of Enclosure and the Ontological Meaning of Mass

1.2.1 Definition of Enclosure

Enclosure refers to a structural state in which the rotational organization of space forms a closed, stable structure under geometric constraints, which cannot freely unfold.

Any enclosure structure must simultaneously satisfy three hard conditions:

1. phase closure (otherwise it cannot exist long-term);
2. rotational continuity (otherwise shear and dissipation occur);
3. channel uniqueness (otherwise phase is not conserved).

These three conditions are not empirical summaries, but necessary conditions for stable existence.

1.2.2 Ontological Definition of Mass (Core)

In the enclosure universe, mass is not a fundamental quantity but an equivalent bookkeeping of enclosure cost:

$$m \equiv \frac{E_{\text{encl}}}{c^2}$$

That is, mass equals the equivalent representation of the minimum energy cost required to maintain an enclosure structure.

This definition immediately yields three conclusions:

- mass is not an "attribute possessed" by an object, but the cost of enclosure;
- different enclosure hierarchies correspond to different mass scales;
- changes in mass are equivalent to changes in enclosure hierarchy.

1.3 Rotation, Magnetism, and the Minimal Magnetic Dipole

1.3.1 Definition of Rotation Ω

Space admits rotational structures, characterized by an angular frequency Ω.

- no rotation → no magnetism;
- rotation without enclosure → propagation state (dynamic magnetism);
- rotation with enclosure → static magnetism.

Magnetism is thus reduced to the geometric manifestation of rotational structure.

1.3.2 Existence Proof of the Minimal Magnetic Dipole (Electron) — Core Derivation Version

Goal of this section: Without introducing any quantum-mechanical postulates, and using only the fixed "template" already locked in Chapter 1 (the three hard enclosure conditions, the definition of mass, $|g| = c$, and the single-channel principle), we prove:

1. the minimal stable enclosure state is necessarily the minimal magnetic dipole;
2. its characteristic rotational frequency (Ω_e) is a structural quantity (defined first, used later);
3. Planck's constant ($\hbar$), as the "enclosure cost per radian," is closed at the electron level:

$$\boxed{\hbar = \frac{m_e c^2}{\Omega_e}}$$

(I) Calling the fixed template: mass–enclosure and the three hard stability conditions

This book has defined mass as the equivalent bookkeeping of enclosure cost: $m \equiv \frac{E_{\text{encl}}}{c^2}$,

therefore for any stable enclosure state, $E_{\text{encl}} = mc^2$.

At the same time, any stable enclosure structure must satisfy the three hard conditions:

- phase closure,
- rotational continuity,
- channel uniqueness.

These are not empirical rules, but necessary conditions for long-term existence.

(II) Define Ω_e first: it is a structural quantity, not introduced ad hoc for $\hbar$

Consider the "smallest possible stable enclosure structure." In this book, "minimal" means:

- it must be closed (otherwise it is not an enclosure state);
- it must have rotational continuity (otherwise shear dissipation prevents long-term stability);
- it must expose only one external channel (otherwise phase cannot be conserved and instability occurs).

Because it must satisfy rotational continuity, the structure must possess an internal rotational rhythm that can be characterized by an angular frequency. For the minimal enclosure state, this angular frequency is not a free family of parameters; it is locked as the unique stable rotational frequency allowed by the combined constraints "closure + continuity + single channel." Hence we give the strict definition:

Ω_e = the characteristic angular frequency of the electron as the minimal stable enclosure state.

Logical status: Ω_e appears prior to $\hbar$. It is defined by the existence of the minimal stable enclosure state and is therefore a structural quantity.

(III) The hard link between Ω_e and $|\mathbf{g}| = c$: rotation frequency from helical-angle allocation

In Section 1.6, this book has fixed the local coordinate system and velocity decomposition, and has locked the unique kinematic constraint: $|\mathbf{g}| = c$

In the local XYZ convention,

$$\mathbf{g} = c \sin\theta \, (\cos\psi \, \hat{\mathbf{x}} + \sin\psi \, \hat{\mathbf{y}}) + c \cos\theta \, \hat{\mathbf{z}}$$

where θ is the helical tilt angle and ψ is the transverse phase angle.

For a "rotational structure," what truly produces magnetism is not the axial component $c\cos\theta$, but the transverse rotational component $g_\perp = c\sin\theta$.
Interpreting this transverse component as angular motion around a local rotational radius r_e (this is precisely the geometric meaning of "rotational continuity"), the angular frequency is defined purely kinematically as

$$\Omega_e \equiv \frac{g_\perp}{r_e} = \frac{c\sin\theta_e}{r_e}$$

This is not a new assumption; it is the geometric definition "angular frequency = tangential speed / radius." The quantities θ_e and r_e are locked by the minimal enclosure closure condition (minimal closure, single channel, no multiple interfaces).
Thus, Ω_e is explicitly grounded in the book's unified constraint$|\mathbf{g}| = c$ and the helical-angle allocation rule, avoiding any "arbitrary introduction of Ω_e."

(IV) Energy bookkeeping for rotational enclosure: define $\hbar$ as the 'enclosure cost per radian'
The ontological definition of an enclosure state is that a rotational structure is forced by geometric constraints to close into a stable structure. Since rotation is periodic, the minimal divisible unit of enclosure energy bookkeeping can only accumulate linearly with the rotational phase. Therefore, for any rotational enclosure state, the enclosure energy must be expressible as $E_{\text{encl}} = \hbar\Omega$, where the physical meaning of $\hbar$ is fixed in this framework as follows:
$\hbar$ = the energy scale corresponding to unit angular frequency, i.e. the 'enclosure cost per radian' (action scale).
This is a bookkeeping relation for enclosure states, not a quantum-mechanical postulate; it relies only on the structural fact that enclosure energy is measured by rotational rhythm.

(V) Closure at the electron level: derive $\hbar = \frac{m_e c^2}{\Omega_e}$ (core result)

For the electron as the minimal stable enclosure state, the enclosure energy is exactly the enclosure cost corresponding to its rest mass: $E_{\text{encl}}^{(e)} = m_e c^2$.

On the other hand, by the rotational-enclosure bookkeeping relation, $E_{\text{encl}}^{(e)} = \hbar\Omega_e$.

These two expressions refer to the same physical quantity $E_{\text{encl}}^{(e)}$, hence they must be equal:

$$\hbar\,\Omega_e = m_e c^2 \Rightarrow \hbar = \frac{m_e c^2}{\Omega_e}.$$

Conclusion: In this system, Planck's constant is no longer an independent natural constant; it is the action scale obtained by closure from "the electron as the minimal stable enclosure state."

Calculation Box 1-1: Auditable closure chain

1. mass definition (book template): $E_{\text{encl}} = mc^2$;
2. electron level: $E_{\text{encl}}^{(e)} = m_e c^2$;
3. rotational enclosure bookkeeping: $E_{\text{encl}}^{(e)} = \hbar\Omega_e$;

4. structural definition of rotation frequency: $\Omega_e = \frac{c \sin\theta_e}{r_e}$ (from the helical-angle allocation under $|\mathbf{g}| = c$);
5. closure: $\boxed{\hbar = \frac{m_e c^2}{\Omega_e}}$

(VI) The minimal magnetic dipole conclusion (automatic consequence of the derivation)
As the minimal stable enclosure state, the electron must contain an enclosed rotational structure. The external manifestation of an enclosed rotation is precisely a static magnetic structure (a closed magnetic flux tube / single-channel manifestation). Therefore, the electron is not "analogous to a magnetic dipole"; rather:

Electron = Minimal rotational enclosure state = The minimal magnetic dipole in nature

At this point, the "minimal magnetic dipole (electron)" and the "structural origin of $\hbar$" are simultaneously locked within the same closure chain.

1.4 The Single-Channel Theorem: Stable Enclosure Allows Only One Interface

1.4.1 Physical Meaning of the Single Channel

The single channel is not an assumption; it is a necessary consequence of enclosure stability.
If an enclosure structure exposes two or more independent interfaces, then inevitably:

- the phase cannot be closed;
- the rotation sense cannot remain continuous;
- and the enclosure must become unstable.

Therefore, we obtain the theorem:
Any stable enclosure state can expose only one external channel.

1.4.2 Single Channel ≡ Closed Magnetic Flux Tube (Core Equivalence)

The external manifestation of a rotational enclosure structure in space necessarily forms:

- an emission port;
- a bypass (circulation) path;
- an absorption port.

This geometric structure is exactly a closed magnetic flux tube.
Hence the second core equivalence of this book follows:
Single channel = Closed magnetic flux tube

1.5 The Three-State Energy Ledger: Enclosure, Propagation, and Re-Enclosure

1.5.1 Definition of the Three States

State	Structure	Energy Scale
Enclosure state	Closed magnetic flux tube	$E_{\text{encl}} = mc^2$
Propagation state	De-enclosed (decoupled) flux tube	$E = \hbar\omega$
Re-enclosure state	Endpoint: enclosure succeeds as a whole	Energy quantization

1.5.2 Upper Limit of Radiative Energy (Core)

The "mother energy" of the electron's flux tube is: $E_0 = m_e c^2 = 511 keV$

Normal electromagnetic radiation satisfies: $E_\gamma \ll m_e c^2$

More precisely, the natural upper bound of strong radiation is: $E_{rad,max} \sim \frac{\alpha}{\pi} m_e c^2$

where:

- α is the single-channel visibility (manifestation) factor;
- π is the geometric closure normalization factor.

1.6 Alignment of the Three-Field Directions with XYZ, Constant Composite Light Speed, and Helical-Angle Allocation (Coordinate and Notation Standard)

To avoid ambiguities in the directions of the three fields and the velocity decomposition in later chapters, we provide here a unified specification of the coordinate system, directions, and helical-geometry notation used throughout this book. The conventions below introduce no new physical assumptions; they serve only to fix the descriptive and computational standard.

At any point, define a local Cartesian coordinate system ((X,Y,Z)), and impose:

- Z axis: $\mathbf{B} \parallel \hat{\mathbf{z}}$ (magnetic field, helical principal axis);
- X axis: $\mathbf{E} \parallel \hat{\mathbf{x}}$;
- Y axis: $\mathbf{A}_g \parallel \hat{\mathbf{y}}$.

Thus the three-field relation is fixed as:

$$\mathbf{B} \parallel \hat{\mathbf{z}}, \quad \mathbf{E} \parallel \hat{\mathbf{x}}, \quad \mathbf{A}_g \parallel \hat{\mathbf{y}}, \quad \mathbf{E} \perp \mathbf{A}_g \perp \mathbf{B}.$$

The magnitude of the spatial velocity field $\mathbf{g}$ satisfies the single axiom: $\boxed{|\mathbf{g}| = c}$

Decompose: $\mathbf{g} = g_x \hat{\mathbf{x}} + g_y \hat{\mathbf{y}} + g_z \hat{\mathbf{z}}, \quad g_x^2 + g_y^2 + g_z^2 = c^2$

Introduce the tilt angle θ and phase angle ψ:

$$g_z = c\cos\theta\,, \quad g_x = c\sin\theta\cos\psi\,, \quad g_y = c\sin\theta\sin\psi$$

so that:

$$\mathbf{g} = c\sin\theta\,(\cos\psi\ \hat{\mathbf{x}} + \sin\psi\ \hat{\mathbf{y}}) + c\cos\theta\ \hat{\mathbf{z}}$$

For a cylindrical helix, the pitch p and radius r satisfy: $\tan\theta = \frac{2\pi r}{p}, \quad p = 2\pi r\ \cot\theta$

This paper uniformly adopts right-handedness: increasing ψ corresponds to the transverse component rotating in the XY plane according to the right-hand rule, while advancing along +Z. The XYZ alignment, the constant composite light-speed constraint, and the helical-angle allocation specified here will serve as the default coordinate and geometric standard for all subsequent chapters and will not be repeated.

1.7 Where Quantization Appears (Key Conclusion)

In the propagation state:

- evolution is continuous;
- interference is possible;
- superposition is possible.

Quantization does not appear in the propagation stage. Quantization appears only at re-enclosure endpoints, where the only meaningful question is whether the structure succeeds as a whole. Therefore, $E = \hbar\omega$ is not a propagation law, but the minimal unit of re-enclosable structure.

1.8 Chapter Summary (Locking the Computational Template)

This chapter establishes the non-negotiable computational template of the entire book:

1. Mass = the energy cost of enclosure;
2. The electron is the minimal magnetic dipole;
3. Single channel ≡ closed magnetic flux tube;
4. $\hbar$ originates from the electron's enclosure structure;
5. Radiation arises from flux-tube decoupling;
6. Quantization appears at re-enclosure endpoints;
7. The three-field directions, the constant composite light speed, and the helical-angle allocation are the unified standard throughout the book.

All later chapters on electrons, protons, atoms, nuclei, electromagnetic waves, photons, quantum phenomena, and applications are only developments of this locked template at different hierarchical levels, rather than new assumptions.

Chapter 2 Electron: The Smallest Magnetic Dipole in Nature, the Closed Magnetic Flux Tube, and the Ontological Origin of the Quantum Energy Scale

2.1 Ontological Positioning: The Electron Is One of the Only Three Stable Enclosure "Building Blocks"

Within the enclosure-universe framework, the electron, proton, and neutron are the only three enclosure states in nature that can remain stable over long timescales and repeatedly participate in structural construction. Among them, the electron plays the most fundamental role: the electron is the smallest rotational enclosure structure, i.e., the "smallest magnetic dipole" in nature. This conclusion is not an analogy; it follows directly from enclosure stability plus the single-channel theorem.

2.2 The Rotational Structure of the Electron: Necessity of Double-Helical Enclosure

2.2.1 Why the Electron Must Rotate

Stable enclosure requires:

1. phase closure;
2. rotational continuity;
3. a unique external channel.

A non-rotational structure cannot produce magnetism or a stable channel. Therefore, the minimal enclosure must be rotational.

2.2.2 Why the Minimal Stable Rotation Must Be "Double-Helical"

A single-loop rotation in three-dimensional space cannot satisfy "closure + continuity" simultaneously.

A double helix is the minimal stable solution that satisfies the three conditions above. Therefore: the electron's ontological structure must be a double-helical rotational enclosure.

2.3 Physical Realization of the Single-Channel Theorem at the Electron Level

2.3.1 The Single Channel Is Not an Assumption

If a stable enclosure exposes two or more independent interfaces, it produces phase multivaluedness and breaks rotational continuity, so the enclosure becomes unstable immediately.

Thus, a stable enclosure state can expose only one external channel.

2.3.2 Single Channel ≡ Closed Magnetic Flux Tube

The external manifestation of the electron's double-helical rotation in space necessarily forms:

- an emission port (externalization of rotation);
- a bypass (circulation) path (topological protection);
- an absorption port (recovery of rotation).

Therefore, at the electron level there is a strict equivalence: Single channel = The electron's closed magnetic flux tube

2.4 The Ontological Basis of Electron Magnetism: The Minimal Magnetic Dipole

Since the electron is both a rotational enclosure state and exposes only one closed magnetic flux tube externally, it follows that:

the electron intrinsically carries static magnetism and is the smallest indivisible magnet in nature. The electron's magnetic moment is not an adjustable parameter, but an external manifestation of enclosure geometry. Its orientation allows only two stable states (aligned/opposed), corresponding to spin 1/2.

2.5 Enclosure Energy of the Electron Flux Tube (Hard-Core Calculation)

This section provides the most critical "mother-energy" calculation of the entire book.

Calculation Box 2-1: Minimal Enclosure Energy of the Electron Flux Tube

As the minimal enclosure state, the electron satisfies: $E_{e,\text{encl}} = \hbar\Omega_e$

Experimentally, the electron rest energy is: $m_e c^2 = 0.511 MeV$

Thus we obtain the enclosure mother relation: $\boxed{\hbar = \frac{m_e c^2}{\Omega_e}}$

Conclusion (auditable): $E_{e,\text{encl}} = m_e c^2 = 0.511$,MeV.

Physical meaning:

- this is not an "additional magnetic energy," but the total enclosure cost required for the electron's existence;
- therefore, radiation does not consume the electron's existence energy.

2.6 Why Radiation Does Not Exhaust the Electron: The Three-State Structure

Electron-related phenomena are strictly divided into three states:

State	Structure	Energy Scale
Enclosure state	Closed magnetic flux tube	$m_e c^2$
Propagation state	Decoupled flux tube	$\hbar\omega$
Re-enclosure state	Endpoint: succeeds as a whole	Energy quantization

Radiation involves only the decoupling of the outer layer of the flux tube and does not touch the enclosure "mother core."

2.7 Magnitude of Normal Electromagnetic Radiation Energy (Hard-Core Ratio)

Calculation Box 2-2: Ratio of Radiative Energy to the Electron Mother Energy

For any electromagnetic radiation: $\frac{E_\gamma}{m_e c^2} = \frac{\hbar\omega}{m_e c^2} = \frac{\omega}{\Omega_e}$

Example (visible light): take $E_\gamma \approx 2\,eV$: $\frac{2}{511000} \approx 3.9 \times 10^{-6}$

Conclusions:

- normal radiation is only at the level of one-millionth of the mother energy;
- this explains why the electron can radiate repeatedly without being exhausted.

2.8 Natural Upper Bound of Strong Radiation (α/π Scale)

Calculation Box 2-3: Upper Bound of Strong Radiation

From the single-channel visibility factor and the geometric closure factor: $\boxed{E_{rad,max} \sim \frac{\alpha}{\pi} m_e c^2}$

Substituting numbers: $\frac{1}{137\pi} \times 511\,keV \approx 1.19\,keV$

Interpretation:

- ultraviolet / soft X-rays lie near this upper bound;
- at higher energies, the enclosure threshold is approached and the process enters a non-"normal electromagnetic radiation" regime.

2.9 Where Quantization Appears (Endpoint Criterion)

- Propagation stage: continuous, interferable, superposable;
- Re-enclosure endpoint: the structure can only succeed or fail as a whole.

Therefore, $\boxed{E = \hbar\omega}$ is not a propagation law, but the minimal unit of re-enclosable structure.

2.10 Ontological Clarification of the "Particle Zoo": Why Only Three Stable Building Blocks Exist

In the preceding chapters, we have rigorously proven that within the unified framework of "enclosure–propagation–re-enclosure," only three enclosure states can remain stable over long timescales and repeatedly participate in structural construction:

- electron (minimal rotational enclosure state);
- proton (second stable enclosure level, higher-order single-channel enclosure);
- neutron (stable enclosure state with a phase-sign compensation shell).

A natural question follows: if these three are the only basic "building blocks" of nature, what are the many "new particles" observed in high-energy physics experiments (e.g., colliders)? This section provides a unified, non-evasive, non-rhetorical answer.

2.10.1 Key Distinction: Stable Enclosure States vs. Transient Background-Flow States

The ontological core of this book is that the basic existents in nature are only two: space and objects; an object appears only when space forms a stable enclosure.

This immediately yields a fundamental distinction:

- Stable enclosure states: satisfy phase closure, rotational continuity, and single-channel uniqueness; persist long-term and can repeatedly build larger structures;
- Non-enclosure / transient states: violate one or more of the three conditions; appear briefly only under specific excitations, then necessarily de-enclose and return to the background flow.

Electrons, protons, and neutrons belong to the first class; the overwhelming majority of "particles" produced in high-energy collisions belong strictly to the second.

2.10.2 What High-Energy Collisions Really Do: Not "Making New Bricks," but "Exciting the Background Flow"

In this system, high-energy particle collisions do not mean "converting energy into new fundamental particles."

They mean that, within extremely short times and extremely small spatial regions, the light-speed double-helical background flow of space is violently perturbed, temporarily departing from stable enclosure conditions.

Such perturbations cause:

- the local rotation sense to be strongly distorted;
- phase closure to be temporarily broken;
- the single-channel condition to be "opened" in the propagation regime.

As a result, many transient structural modes appear in the spatial background flow, recorded in detectors as a series of "new particle events." These are not new "bricks," but transient structural projections of the background flow under strong excitation.

2.10.3 Unified Classification Principle for the "Particle Zoo"

Within this framework, the particles observed in colliders can be unified into three categories:

(I) Stable enclosure building blocks

- electron, proton, neutron;
- possess complete enclosure structures;
- can persist long after experiments;
- constitute the basic skeleton of atoms, nuclei, and the material world.

(II) Metastable enclosure states / resonance states

- appear when enclosure conditions are approached but not fully satisfied;
- extremely short lifetimes;
- typically manifest as "resonance peaks" or "short-lived particles";
- essentially intermediate states in the enclosure process, not independent ontologies.

(III) Pure propagation states (background-flow excitations)

- do not satisfy enclosure conditions;
- manifest only as propagation of perturbations in rotation sense, phase, or density;
- recorded in detectors as "tracks" or "energy deposition";
- ontologically are propagation modes of space flow, not objects.

The so-called "particle zoo" is mainly composed of the latter two categories.

2.10.4 Ontological Repositioning of Fermions and Bosons

In conventional quantum field theory, "fermions/bosons" are treated as two kinds of fundamental particles. In this system, the distinction is repositioned as structural, not ontological:

Fermions

- correspond to stable or near-stable enclosure states;
- possess an enclosure core;
- single channel is unique;
- spin 1/2 originates from the geometric topology of the minimal double-helical enclosure;
- Pauli exclusion is not a "statistical rule," but the structural fact that a single channel cannot be multiply occupied.

Electrons, protons, and neutrons belong to this class.

Bosons

- correspond to non-enclosure or already de-enclosed propagation states;
- do not constitute independent objects;
- are propagation forms of perturbations in rotation sense, phase, or energy;
- can superpose, remain coherent, and occupy the same propagation mode.

Electromagnetic waves, photons, and many "bosonic signals" in collider experiments all belong to the propagation-state manifestations in this system.

2.10.5 Why High-Energy Physics "Sees Many," but Nature "Keeps Only Three"

This is not a contradiction, but a difference in hierarchy:

- high-energy experiments probe the transient structural spectrum of the background flow under extreme excitation;
- under normal conditions, nature preserves only structures that can exist long-term under stable enclosure conditions.

In one sentence: colliders display what kinds of "ripples" space can form, while nature retains only structures that can be stably enclosed as "bricks."

2.10.6 Conclusion of This Section

The "particle zoo" observed in high-energy physics does not imply that nature contains many fundamental particles. These signals mainly correspond to transient structures and propagation modes formed in the background double-helical flow under strong excitation.

Within the unified framework of "enclosure–propagation–re-enclosure," only the three stable enclosure states—electron, proton, and neutron—constitute the basic building blocks of all things; all other "particles" are transient manifestations of the background flow, not independent ontologies.

2.11 Chapter Summary (Final Form of the Electron Chapter)

The electron is one of the only three stable enclosure building blocks in nature. Its ontology is the minimal double-helical rotational enclosure state. This rotation naturally generates a unique closed magnetic flux tube, which serves as the electron's external single-channel interface. The electron is therefore the minimal magnetic dipole, and the enclosure energy of its flux tube equals the electron rest energy $m_e c^2$. Normal electromagnetic radiation involves only the decoupling of the outer layer of the flux tube, at energies far below the enclosure mother core; quantization appears at re-enclosure endpoints rather than during propagation.

Chapter 3 Proton: The Second Stable Enclosure Level, the Unique Dimensionless Combination, and the Starting Point of the Composite Main Channel

3.1 The Theoretical Status of the Proton: The Second Stable Level of Existence

In Chapter 1, we have established:

- stable objects = stable enclosure states of space;
- enclosure states must satisfy: phase closure, rotational continuity, and single-channel uniqueness;
- the electron is the minimal enclosure solution that satisfies these conditions.

Therefore, an unavoidable question arises: above the electron, is a second stable enclosure structure—of significantly larger mass—still allowed to exist?

This chapter will show that if stable enclosure structures are required to remain consistent in

geometry, axiomatics, and dimensional coherence, then the second stable level of existence is not only allowed, but its mass scale is almost uniquely fixed.

3.2 The Proton Is Not a Scaled-Up Electron: Necessity of Higher-Order Enclosure

The electron enclosure state corresponds to the minimal closed unit of rotational closure.
If one attempts to obtain a larger-mass enclosure simply by scaling up this structure, at least one stability condition must be violated:

- phase closure becomes multivalued;
- rotational continuity generates shear;
- multiple interfaces cause the single-channel condition to fail.

Hence, the second stable enclosure level must satisfy: it is not a simple scale enlargement, but a "higher-order enclosure closure."
This implies:

- introducing a new geometric closure factor;
- while simultaneously preserving single-channel uniqueness.

3.3 The Unique Dimensionless Combination for the Second Stable Enclosure Level

3.3.1 Formalizing the Problem

Let the mass of the second stable enclosure level be m_2, and the electron mass be m_e.
Then the ratio m_2/m_e must be dimensionless and must be constructed only from the basic dimensionless constants already appearing in Chapter 1.
Within this system, the available dimensionless candidates are only:

- α: the single-channel visibility (manifestation) factor (from the electromagnetic layer);
- π: the geometric closure normalization factor.

It is not permitted to introduce:

- new coupling constants;
- arbitrary integer multiples;
- ratios of propagation scales.

3.3.2 Uniqueness Theorem (Key Upgrade Point)

Theorem (Uniqueness).
Under the following conditions:

1. the uniqueness of the single channel is not violated;
2. the propagation layer ($\hbar\omega$) is not contaminated;
3. the new enclosure level appears only at an "existence threshold";

the mass ratio of the second stable enclosure level can only take the form: $\frac{m_2}{m_e} \propto \frac{1}{\alpha^2\pi^2}$

The reasons are exclusive:

- α can only appear with an even power (a first power corresponds to visibility, not a stable existence threshold);
- π must appear as a closure power (a single π corresponds only to a single winding and is insufficient to generate a second stable level);
- any combination $\alpha^n\pi^m$ with $n \neq 2$ or $m \neq 2$ either falls into the propagation layer or breaks the single-channel condition.

Therefore, $\alpha^2\pi^2$ is the only admissible combination.

3.3.3 Minimal-Crossing Theorem for Single-Channel Enclosure (8-Crossing Legality Lemma)

In the preceding derivation, the rotational-alignment correction factor of the proton relative to the background helical flow has been written as $\eta_\Omega = \cos(\theta_0/2)$, where θ_0 denotes the effective helical angle at the boundary layer of the secondary enclosure.

This section will show that θ_0 is locked near $30°$, not as a parameter choice but as the minimal topological requirement for single-channel enclosure legality.

Theorem 3.3 (Minimal crossing number for single-channel closure).

For an enclosure structure satisfying the following four constraints:

1. single-channel uniqueness (no external branching interfaces);
2. right-handed single-rotation stability (forbidding left-handed Ω modes);
3. phase single-valuedness (no multivalued ϕ along a closed path);
4. a two-branch structure (coexistence of ϕ positive/negative branches or compensation layers);

the irreducible geometric crossing number $N_\times$ of the closed loop over one full period must satisfy $N_\times = 4k, \quad k \in \mathbb{Z}^+$, and the minimal realizable value is $\boxed{N_\times = 8}$.

Proof (legality by contradiction).

- If $N_\times = 4$, then phase multivaluedness or rotation reversal necessarily occurs at the single-channel junction, violating constraints (2) and (3);
- If $N_\times = 6$, the two branches cannot complete closure simultaneously in the homotopy sense, violating constraint (4);
- Only when $N_\times = 8$ can all four constraints be satisfied simultaneously, without geometric degeneration or topological splitting.

Thus, "8 crossings" are not an empirical input but the minimal topological legality value for single-channel enclosure.

Corollary 3.3.1 (Geometric origin of the rotational-alignment correction).

From $N_\times = 8$, one obtains the effective helical angle $\theta_0 \approx 30°$, hence $\eta_\Omega = \cos(\theta_0/2) \approx \cos 15° \approx 0.966$. This value is a consequence of geometric legality, not a posterior correction introduced to fit the mass ratio.

Role of this subsection: This subsection provides a topological origin for the rotational-alignment factor in the proton–electron mass ratio, removing potential suspicions of "fine-tuning" or "number fitting."

3.4 Rotational-Alignment Correction Factor η_Ω in Higher-Order Enclosure: Background Double Helix, Single-Channel Reconnection, and Geometric Necessity

In the previous section, enclosure stability and dimensional consistency imply that the main mass scale of the second stable enclosure level (the proton) is uniquely fixed (dimensionlessly) as $\frac{m_p}{m_e} \sim \frac{1}{\alpha^2\pi^2}$. However, comparing this main scale with the experimentally measured proton–electron mass ratio yields a systematic deviation of about 4%.

This section proves that the deviation is neither a model defect nor an after-the-fact correction, but an unavoidable geometric result when a higher-order enclosure state is embedded in a light-speed double-helical background.

3.4.1 The Background Double Helix as an Unremovable External Geometric Constraint

The single axiom of this book states that for any stable object at rest relative to the observer, the surrounding space performs a right-handed double-helical divergent motion at light speed c. This background flow has three irreducible geometric features:

1. a globally right-handed rotation sense;
2. rotation is not planar but a three-dimensional helix with axial advance;
3. locally, the background double helix defines a preferred rotation axis for any enclosure state.

Hence any local enclosure state (electron or proton) does not rotate in "static space," but is embedded within a double-helical background that is already rotating and oriented.

3.4.2 Why No Rotational-Alignment Correction Appears at the Electron Level

The electron is the minimal stable enclosure state. Its structure contains only one minimal double-helical closed unit, and its rotation axis can locally align perfectly with the background double-helix axis.

At the electron level:

- there is no cooperative splicing of multiple rotation segments;
- there is no path foldback;
- no single-channel reconnection is required among different rotational branches.

Therefore, electron rotation may be treated as ideal co-rotation with no systematic loss of rotational alignment. This is precisely why no η_Ω correction is introduced at the electron level.

3.4.3 Essential Difference of the Proton Enclosure State: Higher-Order Enclosure under the Single-Channel Constraint

The proton is not a simple scaled-up electron but a higher-order stable enclosure state. Its structural features include at least:

1. cooperative closure of multiple rotational enclosure units within a finite volume;
2. foldback and reconnection of enclosure paths;
3. only one external single-channel interface is still allowed (the single-channel theorem cannot be violated).

Together, these imply a key geometric fact: the proton's internal rotational structure cannot achieve global co-rotation with the background double helix across the entire enclosure volume. This is not dynamical insufficiency but geometric incompatibility.

3.4.4 Three Irreducible Geometric Sources of Rotational-Alignment Breaking

In higher-order enclosure, the following three geometric effects necessarily appear and cannot be eliminated by structural adjustment:

(1) Local deflection of the rotation axis (orthogonality effect).

Path foldback inevitably deflects the local rotation axis from the background axis, projecting part of the rotation component into orthogonal directions, thus reducing effective co-rotation.

(2) Pitch mismatch and accumulated torsion (torsion effect).

To satisfy phase closure, different path segments must adopt slightly different effective pitches; after multi-segment accumulation, torsion errors cannot fully cancel, lowering the average axial co-rotation efficiency.

(3) Irreversible shear induced by single-channel reconnection.

Multiple internal rotational branches must ultimately merge externally into the same emission–bypass–absorption structure. Branches have angular-velocity differences before reconnection;

reconnection necessarily introduces rotational shear, converting part of rotational energy into internal tension required for stability.

3.4.5 Definition of the Rotational-Alignment Correction Factor η_Ω

Combining the effects above, the relation between the effective rotational frequency Ω_{eff} and the ideal co-rotation frequency Ω_{ideal} can only be written as: $\Omega_{\text{eff}} = \eta_\Omega \, \Omega_{\text{ideal}}, \quad 0 < \eta_\Omega < 1.$

Here η_Ω is defined as: the average co-rotation efficiency of a higher-order enclosure state embedded in the background double helix, reduced jointly by axis deflection, pitch torsion, and single-channel shear.

It is a purely geometric, dimensionless, non-empirical correction factor.

3.4.6 Symmetric Torsion in Single-Channel Reconnection and a Semi-Quantitative Expression

In higher-order enclosure, multiple internal helices must merge into a single channel externally. Geometrically, this merging can only be achieved by minimal symmetric torsion:

each local helix is symmetrically tilted relative to the background axis by $\delta = \pm\frac{\theta_0}{2}$,

where θ_0 is the forced helical angle of the minimal stable double-helical enclosure.

Under this symmetry, the effective co-rotation component equals the average projection of rotation onto the background axis: $\eta_\Omega = \langle \cos\delta \rangle = \cos\left(\frac{\theta_0}{2}\right)$.

3.4.7 Forced Angle of the Minimal Stable Double Helix $\theta_0 \simeq 30^\circ$ (Geometric Locking)

A minimal stable three-dimensional double-helical closed enclosure is not arbitrary. To satisfy simultaneously:

- phase closure;
- rotational continuity;
- single-channel uniqueness;

the enclosure path must complete 8 irreducible geometric crossings within one full period.

Under this condition, transverse rotation and axial advance must contribute at comparable orders to closure; otherwise, crossings become insufficient or redundant and the enclosure becomes unstable.

Combining "8 crossings per period" with the helix geometry $\tan\theta = \frac{2\pi r}{p}$

one obtains the minimal stable solution: $\tan\theta_0 \simeq \frac{1}{\sqrt{3}} \Leftrightarrow \theta_0 \simeq 30^\circ$.

This angle is not chosen; it is enforced by the geometric constraints of the minimal stable double-helical enclosure.

3.4.8 Numerical Closure and the Final Form of the Proton–Electron Mass Ratio

With $\eta_\Omega = \cos\left(\frac{\theta_0}{2}\right)$ and $\theta_0 \simeq 30^\circ$, we obtain $\eta_\Omega = \cos 15^\circ \simeq 0.9659 \approx 0.966$.

Therefore the final expression for the proton–electron mass ratio in this system is: $\frac{m_p}{m_e} \approx \frac{\eta_\Omega}{\alpha^2\pi^2}$,

$\eta_\Omega = \cos\left(\frac{\theta_0}{2}\right) \simeq 0.966.$

Here:

- $\alpha^2\pi^2$ sets the dimensionless existence threshold of the second stable level;

- η_Ω represents the unavoidable geometric co-rotation loss when embedding a higher-order enclosure state into the background double helix.

They originate from different layers and introduce no parameter redundancy.
Summary of this section.
The few-percent deviation of the proton mass relative to the $\alpha^2\pi^2$ main scale is not an empirical correction but an unavoidable projection loss induced by single-channel reconnection within a light-speed double-helical background. The existence of this deviation is direct geometric evidence that the proton is not a simple scaled-up electron.

3.5 The Proton's Single Channel and Closed Magnetic Flux Tube

As the second stable enclosure level, the proton must still satisfy the single-channel theorem. The geometric consequences are:

- the proton exposes only one external interface;
- the external manifestation of this interface is a closed magnetic flux tube;
- the proton participates in all nuclear processes only through this single channel.

Therefore: Proton = A higher-order single-channel closed-flux-tube enclosure structure.
This step locks the "proton" directly to the "composite main channel."

3.6 Structural Proof That $p + p$ Cannot Form a Nucleus

Consider two protons attempting to form a composite enclosure:

- both higher-order enclosures carry strong rotational structures;
- the two single-channel interfaces cannot merge while simultaneously preserving phase closure and rotational continuity;
- the enclosure impedance of the composite main channel exceeds the stability threshold.

Thus: $p + p$ cannot form a stable enclosure state.
This conclusion does not rely on the charge-repulsion hypothesis; it follows purely from enclosure-structure conditions.

3.7 The Proton as the "Starting Point of Nuclear Existence"

This chapter shows that:

- the proton is the second stable level of existence;
- its mass scale is uniquely locked by $\alpha^2\pi^2$;
- its single-channel structure is a necessary interface for nuclear synthesis;
- but protons alone cannot establish a composite main channel.

This provides a strict theoretical necessity for introducing the neutron in the next chapter.

3.8 Chapter Summary

The proton is not a simple scaled-up electron, but the second stable enclosure level permitted by space under the constraints of the single channel, the axioms, and dimensional consistency. Its mass scale is uniquely determined by $\alpha^2\pi^2$, and its closed-flux-tube structure makes it a necessary interface for nuclear synthesis. However, the structural instability of $p + p$ implies that a new enclosure unit must be introduced to reduce the impedance of the composite enclosure—necessitating the neutron.

Chapter 4 Neutron: Necessity of a ϕ-Compensated Enclosure, the Impedance-Reduction Mechanism of the Composite Main Channel, and β-Reconstruction Calculations

4.1 Goal of This Chapter: Deriving "The Neutron Must Exist" from Enclosure-Stability Conditions

Chapter 3 has established two hard conclusions:

1. The proton is the second stable level of existence, satisfying $\frac{m_p}{m_e} \approx \frac{\eta_\Omega}{\alpha^2\pi^2}$ and it exposes externally only a single-channel magnetic flux-tube interface;
2. $p + p$ cannot form a stable enclosed composite main channel (structural instability).

The task of this chapter is as follows: rather than assuming the neutron empirically, we derive from the mathematical stability conditions of composite enclosure that a ϕ-compensated enclosure unit must be introduced—this unit is the neutron. We further rewrite the weak interaction as a reconstruction of the compensation shell and provide auditable Q-value calculations.

4.2 Minimal Energy Functional for the Composite Main Channel (Hard-Core Starting Point)

In the enclosure universe, a composite enclosure state must simultaneously satisfy:

- phase closure;
- rotational continuity;
- single-channel uniqueness;
- $|\mathbf{g}| = c$ (Chapter 1).

For a minimal description of the "in-nucleus composite main channel," we introduce an effective energy functional (retaining only the lowest-order terms, each with clear physical meaning):

$$\mathcal{F}[\phi, \mathbf{g}; n] \int_V \Big(\underbrace{\frac{\kappa_\phi}{2}|\nabla\phi|^2}_{\text{Phase tension}} + \underbrace{\frac{\kappa\Omega}{2}|\nabla \times \mathbf{g}|^2}_{\text{Rotating encapsulation}} + \underbrace{\frac{\kappa_n}{2}|\nabla n|^2}_{\text{enclosure density boundary}} \Big) dV + \underbrace{\mathcal{F}_{core}}_{\text{Core constraint}}$$

where:

- ϕ: single-channel phase (determines charge sign / winding number);
- $\mathbf{g}$: the geometric potential of helical flow (determines rotation $\Omega = |\nabla \times \mathbf{g}|$);
- $n = \nabla \cdot (\mu_g \mathbf{g})$: enclosure density (source of mass/gravity);
- $\kappa_\phi, \kappa_\Omega, \kappa_n > 0$: effective stiffness constants (only positive definiteness is required; no fitting is performed at this stage).

Key point: enclosure stability requires that $\mathcal{F}$ possess a local minimum.

4.3 Why the $p + p$ Composite Must Be Unstable: The Phase Term Produces No Lower Bound

4.3.1 Positive Proton Charge Implies a ϕ-Winding Number of (+1)

The proton's external charge is determined by the number of phase closures (locked in Chapter 3):

$$N_\phi(p) = +1$$

If two protons are combined without introducing an additional compensation layer, then the total

winding number is $N_\phi^{(pp)} = +2$.

4.3.2 A Winding Number of (+2) Necessarily Increases $|\nabla\phi|$ in the Connection Region

In a single-channel structure, a larger winding number means that within a finite region the phase must complete a larger total rotation. Therefore, the phase-tension term obeys a lower-bound estimate:

$$\int_V |\nabla\phi|^2\, dV \gtrsim C\, N_\phi^2,$$

where C is a geometric constant (dependent on nuclear scale).

Hence for $p + p$,

$$\mathcal{F}_\phi^{(pp)} \equiv \frac{\kappa_\phi}{2}\int |\nabla\phi|^2 dV \gtrsim \frac{\kappa_\phi}{2} C\,(2)^2 = 2\kappa_\phi C$$

This means that the phase-conflict energy grows as N_ϕ^2.

When $N_\phi = 2$, this term significantly raises the total energy and induces:

- increased rotational shear (since $|\nabla \times \mathbf{g}|$ must rearrange to "smooth" ϕ);
- a steeper enclosure-density boundary layer (increased $(|\nabla n|)$;
- thus the system lacks a stable local minimum and transitions into de-enclosure (decay / failure of nuclear formation).

This is the mathematical mechanism for why $p + p$ cannot form a nucleus:

$N_\phi^{(pp)} = 2 \Rightarrow \mathcal{F}_\phi$ too large $\Rightarrow \mathcal{F}$ has no stable minimum $\Rightarrow$ no nuclear enclosure

4.4 Stability Implies a "ϕ-Compensation Layer Must Exist"

If two protons are to coexist with more nucleons to form a stable nuclear enclosure, the phase-tension term must be suppressed. The minimal way to do so is not to modify the proton ontology (impossible), but to introduce a structure such that the total winding number is partially canceled at the externally visible level.

That is, there must exist an enclosure unit with phase winding $N_\phi(\text{comp}) = -1$.

Then, in a composite structure containing "proton core + compensation layer," the total winding becomes $N_\phi^{(total)} = +1 + (-1) = 0$.

This is precisely the definition of the neutron: Neutron = Proton enclosure core + ϕ-sign-reversed compensation shell.

It immediately implies:

- external charge $q = 0$ (phase compensation);
- but the rotational enclosure term remains (κ_Ω does not vanish), hence a magnetic moment is still allowed.

4.5 The Strict Meaning of "Impedance Reduction": Making the Functional Admit a Stable Minimum

Introducing the neutron into the composite system converts N_ϕ from "multiple positive-charge superposition" into "local compensation." Mathematically:

$$\mathcal{F}_\phi \propto N_\phi^2 \Rightarrow (pp): N_\phi = 2 vs (pn): N_\phi = 1 \text{or } 0 \text{ (local compensation)}$$

Therefore, $\mathcal{F}_{\phi}^{(pn)} < \mathcal{F}_{\phi}^{(pp)}$, and the induced rearrangement burdens in $\mathcal{F}_{\Omega}$ and $\mathcal{F}_{n}$ are further reduced, so that the total functional regains a local minimum at nuclear scale (stable nuclear existence).

This is the strict mathematical definition of "neutron impedance reduction":

Introduce neutron $\Rightarrow \mathcal{F}$ admits a stable minimum $\Rightarrow$ Nuclear enclosure can exist

4.6 Ontological Interpretation of the Weak Interaction: Reconstruction Dynamics of the Compensation Shell

In this model, weak processes are not a fourth fundamental force. Rather, when the system deviates from the low-impedance valley of enclosure, the ϕ-compensation shell decouples or reconstructs, returning the system to a lower-$\mathcal{F}$ configuration. The minimal reconstruction process is β decay.

4.7 Hard-Core Q-Value Formulae for β Decay (No New Constants Introduced)

Calculation Box 4-1:$\beta^{-}Q$value (general form)

$$Q_{\beta^-}\Delta E_{np} + [B(Z,N) - B(Z+1,N-1)]m_e c^2$$

Calculation Box 4-2: β^{+}/ECQ values (general forms)

$$Q_{\beta^+} - \Delta E_{np} + [B(Z,N) - B(Z-1,N+1)]2m_e c^2 Q_{\mathrm{EC}} - \Delta E_{np} + [B(Z,N) - B(Z-1,N+1)]$$

where:

- $\Delta E_{np} = (m_n - m_p)c^2 \simeq 1.293MeV$;
- $m_e c^2 = 0.511MeV$;
- $B(Z,N)$can be taken directly from AME2020/AME2024 nuclear mass tables.

4.8 β-Decay Example Table (≥3 Auditable Cases)

Calculation Box 4-3: Example 1 — 1—^{3}H→^{3}He (β⁻)

Given:

- $B(^{3}\mathrm{H}) \approx 8.482MeV$
- $B(^{3}\mathrm{He}) \approx 7.718MeV$
- $\Delta B = 0.764MeV$

Substitute: $Q_{\beta^-} = 1.293 - 0.511 - 0.764 = 0.018MeV = 18keV$

Measured value: 18.6 keV (consistent).

Calculation Box 4-4: Example 2 — 2—^{14}C→^{14}N (β⁻)

- $B(^{14}\mathrm{C}) \approx 105.285MeV$
- $B(^{14}\mathrm{N}) \approx 104.659MeV$
- $\Delta B \approx 0.626MeV$

$Q_{\beta^-} = 1.293 - 0.511 - 0.626 = 0.156MeV$

Measured value: 156 keV (consistent; the lifetime depends on matrix-element suppression, which is distinct from the existence criterion given by Q).

Calculation Box 4-5: Example 3 —3—^{22}Na→^{22}Ne （directioncriterionforβ^+/EC）

- $B(^{22}\mathrm{Na}) \approx 189.0MeV$
- $B(^{22}\mathrm{Ne}) \approx 189.6MeV$ （AME data）

EC: $Q_{\mathrm{EC}} \approx -1.293 + 0.6 = -0.693MeV$

β⁺: $Q_{\beta^+} \approx -1.293 + 0.6 - 1.022 = -1.715MeV$

This indicates that the reconstruction path is unfavorable under this simplified estimate. In practice, the dominant branch may also depend on electron shells / excited states, but the directional criterion is given by the above Q formulae.

4.9 Chapter Summary

The existence of the neutron follows directly from the stability conditions of the composite main channel: without introducing a ϕ-compensation shell, the phase-tension term in $p + p$ grows as N_{ϕ}^2, so the enclosure energy functional admits no stable minimum and nuclear formation fails. The neutron, defined as "proton core + ϕ-sign-reversed compensation shell," reduces phase tension and smooths rotational shear, restoring a stable local minimum for the composite main channel (impedance reduction). The weak interaction is not a fourth force but a reconstruction rule for the compensation shell. The Q values of β decay follow from ΔE_{np}, binding-energy differences, and $m_e c^2$, and their directions and magnitudes are consistent with experiment across multiple nuclei.

Chapter 5 Atomic Nuclei: Existence Criteria for the Composite Main Channel, the Stability Band, and an Ontological Calculation of the Strong Nuclear Force

5.1 Goal and Method of This Chapter

Chapters 3 and 4 have completed three key steps:

1. The proton was established as the second stable enclosure level, and $p + p$ was shown to be structurally unstable;
2. The neutron was strictly derived as the third stable enclosure level via ϕ-compensation, whose role is to reduce the enclosure impedance of the composite main channel;
3. The weak interaction was redefined as the reconstruction dynamics of the compensation shell.

The task of this chapter is the following: based on these results, using a unified energy functional and an existence criterion, we derive which nuclei can exist and which cannot, and we provide a strict ontological definition of the strong nuclear force.

5.2 Existence Criterion for the Composite Main Channel (Mathematical Definition)

Consider a nuclear enclosure body composed of Z protons and N neutrons.
Its stability is not determined by the "magnitude of interaction forces," but by whether the composite main channel admits a stable minimum.
Define the effective energy functional for nuclear enclosure (lowest order, preserving positive definiteness):

$$\mathcal{F}_{nuc} \int_V \left(\frac{\kappa_\phi}{2}\left|\nabla\phi\right|^2 + \frac{\kappa_\Omega}{2}\left|\nabla \times \mathbf{g}\right|^2 + \frac{\kappa_n}{2}\left|\nabla n\right|^2\right) dV + \mathcal{F}_{core}(Z, N)$$

where the three terms represent:

- the phase-tension term (cost of charge/phase conflict);
- the rotational-enclosure term (cost of magnetic-flux-tube continuity);
- the density-boundary term (boundary cost due to finite nuclear scale).

Definition (Existence criterion).

If $\mathcal{F}_{nuc}$ possesses a local minimum at nuclear scale, the nucleus stably exists; otherwise, the nucleus does not exist or is extremely unstable.

5.3 Ontological Definition of the Strong Nuclear Force (Not an Interaction)

The existence criterion immediately yields:

Strong nuclear force = The structural constraint that determines whether a multi-nucleon system admits a stable composite main channel.

Therefore:

- the strong nuclear force is not a fundamental interaction;
- "attraction/repulsion" is merely the external manifestation of existence or non-existence;
- the strong nuclear force must be short-ranged and saturating (determined by enclosure geometry).

5.4 Existence of the Deuteron: The Lowest-Order Composite Main Channel (Hard-Core Example)

5.4.1 Special Status of the Deuteron

The deuteron $(p+n)$ is the first permitted composite main channel:

- $p+p$: phase tension $N_\phi^2 = 4$, no minimum;
- $p+n$: ϕ-compensation makes $N_\phi \approx 1$, so a minimum can appear.

5.4.2 Natural Scale of the Deuteron Binding Energy

From the template of Chapter 1, strong nuclear enclosure is allowed only in the "endpoint residual layer," whose natural energy scale is: $B_0 \sim \frac{\alpha}{\pi} m_p c^2$

Substituting values: $\frac{1}{137\pi} \times 938.27MeV \approx 2.18MeV$

Experimental value: $B_d^{exp} = 2.2246tMeV$

Hard-core conclusion:

the deuteron binding energy is not a fitted result, but a natural scale uniquely set by the single-channel visibility factor α and the geometric closure factor π.

5.5 Sign Criterion of Separation Energies: Direct Test of Existence/Non-Existence

Define the neutron and proton separation energies:

$$S_n(Z,N) = B(Z,N) - B(Z,N-1),\ S_p(Z,N) = B(Z,N) - B(Z-1,N)$$

In this model:

- $S_{n,p} > 0$: removing a nucleon increases $\mathcal{F}_{nuc} \rightarrow$ nucleus is stable;
- $S_{n,p} \leq 0$: removing a nucleon decreases $\mathcal{F}_{nuc} \rightarrow$ nucleus does not exist or is transient.

Calculation Box 5-1: Sign Table for Existing/Non-Existing Nuclei (Examples)

Nuclide	Structure	Criterion	Conclusion
^{2}H	$p+n$	$S_n, S_p > 0$	stable
^{4}He	$2p+2n$	$S_n, S_p \gg 0$	stable
^{5}He	$2p+3n$	$S_n < 0$	does not exist
^{8}Be	$4p+4n$	$S_\alpha \approx 0$	extremely

No potential model is required: existence can be judged by sign alone.

5.6 Mathematical Origin of the Stability Band

On the Z–N plane, stable nuclei concentrate along a "stability band."
In this model, the stability band is precisely the trajectory satisfying:

$$\frac{\partial \mathcal{F}_{nuc}}{\partial Z} \approx \frac{\partial \mathcal{F}_{nuc}}{\partial N} \approx 0$$

That is, the balance path between:

- phase tension (increasing with Z);
- compensation-shell cost (increasing with N),

within the composite main channel.
Deviation from this trajectory necessarily triggers:

- β reconstruction (Chapter 4);
- particle emission or fission.

5.7 Short-Range and Saturation of the Strong Nuclear Force (Strict Origin)

5.7.1 Short-Range Character

- $|\nabla\phi|^2, |\nabla \times \mathbf{g}|^2, |\nabla n|^2$ are concentrated at nuclear scale;
- beyond that scale, $\mathcal{F}_{nuc}$ no longer decreases.

Therefore, the strong nuclear force must be short-ranged.

5.7.2 Saturation

As the number of nucleons increases:

- phase and boundary terms scale with volume/surface area;
- compensation efficiency reaches an upper limit.

When adding additional nucleons no longer decreases $\mathcal{F}_{nuc}$, the nucleus reaches saturation.

5.8 Unified Interpretation of Fission and Fusion

- Fusion: multiple small enclosures merge and $\mathcal{F}_{nuc}$ decreases;
- Fission: one large enclosure splits into multiple enclosures with lower $\mathcal{F}_{nuc}$.

Both are choices of pathways that minimize the energy functional.

5.9 Chapter Summary

The existence of atomic nuclei is determined by whether the composite-main-channel energy functional admits a stable local minimum. The strong nuclear force is not a fundamental interaction, but the existence criterion for whether a multi-nucleon system can form a stable enclosure under the constraints of a single channel, phase, and rotation continuity. The MeV-scale binding energy of the deuteron, the sign criterion of separation energies, and the emergence of the stability band all follow directly from this criterion.

Chapter 6 Single-Channel Enclosure and the Shell-Capacity Theorem: From Principal-Rotation Geometry to the Necessity of $2n^2$

6.1 Ontological Definition of the Single Channel: A Geometric Fact Prior to All Counting

Within the unified framework of this book, any stably existing object—whether an electron, proton, or neutron—is not assembled by splicing several "independent rotating parts," but is an enclosure structure organized around a unique principal rotation.

Definition 6.1 (Principal rotation and single channel).

For any stable enclosure state, there exists one and only one principal rotation axis, denoted by Ω. Along this axis, the enclosure structure forms an indivisible global channel in space, called the single channel.

- one end of the channel manifests as an emission port (N pole);
- the other end manifests as an absorption port (S pole);
- inside the channel, branching, parallel channels, or superposed closed loops are not allowed.

Geometric pointing rule (right-hand rule).

To eliminate any ambiguity, a unified geometric pointing convention is adopted throughout the book: grasp the direction of the principal rotation with the right hand; the direction pointed by the right thumb defines the emission-port direction (N pole) of the single channel. This rule is identical for electrons, protons, and neutrons, and does not change with scale or structural complexity.

6.2 Why the Single Channel Must Be Unique: A First-Principles Constraint of Stability

Lemma 6.1 (Uniqueness of the single channel).

If an enclosure structure contains two or more mutually independent rotational channels, the structure must be unstable.

Proof (geometric).

- Multiple channels imply multiple independent rotation axes;
- different rotation axes cannot simultaneously maintain rotational continuity in space;
- at any merging point, phase multivaluedness or rotational discontinuity necessarily arises;
- the result is enclosure breakup and reversion to a propagation state.

Therefore: Stable enclosure ⇔ Single-channel uniqueness.

6.3 Unified Geometric Status of Electron, Proton, and Neutron

It must be emphasized that:

- Electron: the minimal single-channel enclosure state, with a tightly enclosed internal double-helix;
- Proton: a higher-order single-channel enclosure state, containing multi-layer rotational structure, while the principal rotation remains unique;
- Neutron: on the basis of the proton's single channel, a phase-sign-reversed compensation shell is introduced, while the principal rotation remains unchanged.

Hence, electrons, protons, and neutrons are not three different kinds of "spinning objects," but manifestations of the same single-channel geometric prototype at different enclosure levels. This point is the absolute prerequisite for the subsequent shell counting.

6.4 Ontological Repositioning of Atomic Structure: Standing Waves Occur on the Single Channel

Traditional atomic models treat the electron as a "point particle orbiting the nucleus."
In this system, that description is completely replaced by:

- the electron is an endpoint state of a single-channel enclosure;
- the electron distribution in an atom is a set of standing-wave modes on the global single channel.

Therefore, the atomic-structure problem is no longer an "electron orbit problem," but rather: given a single-channel enclosure background, how many standing-wave modes can satisfy phase closure and rotational continuity?

6.5 Radial Closure Order n: The Geometric Origin of the Phase Budget

Definition 6.2 (Radial closure order).
On the single channel, any stable standing-wave mode must satisfy phase single-valuedness:

$$\oint \nabla \phi \cdot d\ell = 2\pi n, \quad n \in \mathbb{Z}^+.$$

Here n denotes the total number of phase windings permitted to complete one full enclosure closure along the single channel, and is called the radial closure order.
It is not a postulated quantum number, but a bookkeeping quantity of enclosure geometry.

6.6 Budget Constraint on Angular Complexity ℓ

Lemma 6.2 (Upper bound on angular complexity).
For a given radial closure order n, the realizable angular complexity ℓ must satisfy: $\boxed{\ell \leq n - 1}$
Proof sketch.

- each additional angular node requires one additional phase winding;
- the total available phase winding number is fixed by n;
- if $\ell \geq n$, the phase budget is exceeded and the standing wave cannot close.

Hence the physically allowed angular complexity is strictly limited to $\ell = 0,1,\ldots,n-1$

6.7 Counting Spatial Modes under Spherical Symmetry

Theorem 6.1 (Dimension of spatial mode space).
Under spherically symmetric boundary conditions, the number of independent spatial standing-wave modes with angular complexity ℓ is: $\boxed{2\ell + 1}$
This is a purely geometric dimension theorem of the spherical-harmonic function space, independent of any quantum-mechanical operators or postulates.

6.8 Two-Orientation Stability under the Single Channel (Geometric Origin of Spin)

Theorem 6.2 (Two-orientation stability).
Under the single-channel constraint, any given spatial standing-wave mode admits at most two stable enclosure orientations.
Geometric interpretation.

- along the principal rotation, only two complementary orientations exist: co-rotation and counter-rotation;
- introducing a third independent orientation breaks rotational continuity;

- single-channel uniqueness is violated and the structure de-encloses.

These two orientations are the geometric origin of spin, rather than an independent quantum degree of freedom.

6.9 Rigorous Derivation of the Shell-Capacity Theorem $2n^2$

Combining the above results:

- $\ell = 0,1,\dots,n-1$;
- each ℓ corresponds to $2\ell+1$ spatial modes;
- each mode allows 2 orientations;

the maximal capacity of the n-th shell is: $N(n)\& = 2\sum_{\ell=0}^{n-1}(2\ell+1)\& = 2n^2$.

6.10 Geometric Origin of the Periodic Table

The capacity sequence 2,8,18,32, ... is not an empirical rule, but a geometric invariant of the standing-wave mode-space dimension under a single-channel enclosure background.

The complexity of filling order affects only "which mode is occupied first," and cannot change "how many modes exist."

6.11 Chapter Summary (Unifying Concepts and Counting)

Starting from the first-principles geometric fact of "single channel—principal rotation," this chapter provides a unified explanation of the rotational structures of electrons, protons, and neutrons, and on this basis rigorously derives the shell-capacity theorem $2n^2$. Spin, shells, and the periodic table are not independent assumptions, but direct consequences of single-channel enclosure geometry.

Chapter 7 From Atoms to Electromagnetism: Flux-Tube Decoupling, Transitions, and Criteria for Dynamic Magnetism

7.1 Positioning of This Chapter: The "Only Physical Channel" Between Atoms and Electromagnetic Waves

In the first six chapters, we have completed three tasks:

1. established the electron, proton, and neutron as the only three stable enclosure building blocks;
2. proved that an atom = nuclear single channel + electronic single channel → a global single-channel standing wave;
3. identified the structural origin of atomic stability, energy levels, and periodicity.

This chapter addresses the following problem: when an atom undergoes a transition, what structurally "actually happens," and when are electromagnetic waves produced?

The answer is not "the electron drops an orbit," but that the local enclosure of the electron's magnetic flux tube is forced to decouple, forming a propagating dynamic-magnetic structure.

7.2 Ontology of Transitions: Rearrangement of the Standing-Wave Structure Rather Than Particle Motion

7.2.1 What Are the Objects Before and After a Transition?

- Before the transition:
    - the propagation state of the electron flux tube;
    - a stable standing wave formed within the nucleus–electron global single channel.
- After the transition:
    - the original standing-wave mode no longer satisfies phase closure;
    - the system must search for a new realizable standing wave.

Key point: the electron ontology (the small double-helical ring) always remains enclosed and does not disintegrate.

7.2.2 Single-Channel Compression Theorem for Heavy Nuclei: From Multiple Nucleons to One External Interface

Earlier, the single-channel principle was proven to be a necessary condition for stable enclosure. However, for heavy nuclei containing many nucleons (e.g., Pb-208), a key question must be addressed:

How can multiple nucleons preserve their own internal enclosures while the nucleus presents externally only a single-channel interface?

This subsection gives a strict hierarchical answer.

Theorem 7.1 (Nuclear-level single-channel compression theorem).

In an atomic nucleus dominated by strong enclosure density n:

- each nucleon internally maintains its nucleon-level single-channel enclosure;
- but at the nuclear scale, all nucleon channels undergo phase averaging and compensatory compression in angular degrees of freedom;
- therefore, the nucleus as a whole allows externally only one effective single-channel interface.

This interface defines the effective principal axis of the nucleus.

Proof (hierarchical separation).

1. Internal level: the nucleon-level single channel maintains each nucleon's mass and local stability, acting only on the fm scale;
2. Nuclear level: strong n-enclosure forces angular modes to be averaged, and all directions except the minimum-cost direction are exponentially suppressed;
3. External interface: ϕ-readout, propagation coupling, and re-enclosure discrimination occur only along the minimum-cost direction, forming a unique effective principal axis.

Thus, "internal multi-channel" and "external single-channel" are not contradictory but are a necessary result of hierarchical division of labor.

Corollary 7.1 (Nuclear magnetic moment and directionality).

That a nuclear magnetic moment can be represented by a single direction is not accidental; it is a direct manifestation of the nuclear-level effective principal axis. Nuclear internal complexity enters the visible layer only through residual forms (hyperfine structure, isotope shifts).

Role of this subsection.

This clarifies how the single-channel principle applies to heavy nuclei, avoiding the misinterpretation that a "rigid single channel" implies "all internal degrees of freedom must vanish," and providing a structural basis for later discussions of atomic spectra, magnetic moments, and anisotropy.

7.3 Where Decoupling Occurs: The Outer Layer of the Flux Tube, Not the Enclosure Mother Core

Review of Enclosure Hierarchy (Quantitative)

Mother enclosure energy of the electron flux tube: $E_{mother} = m_e c^2 = 0.511 MeV$

Typical atomic transition energy: $E_{atom} \sim 1 - 10 eV$

Ratio: $\frac{E_{atom}}{E_{mother}} \sim 10^{-6} - 10^{-5}$

Conclusion: a transition can only occur in the weakly enclosed outer region of the flux tube.

7.4 Necessary and Sufficient Conditions for Dynamic Magnetism (Computable Criteria)

7.4.1 Necessary Condition: The Standing Wave Cannot Be Locally Rearranged

Let the standing-wave energy difference before and after transition be: $\Delta E = E_i - E_f$

If this energy cannot be absorbed by a new locally enclosed standing wave, then decoupling must occur.

7.4.2 Sufficient Condition: Energy Exceeds the Local Enclosure Threshold

The natural threshold of local enclosure is set by the visibility layer: $E_{local,max} \sim \frac{\alpha}{\pi} m_e c^2$

Substituting values: $\frac{1}{137\pi} \times 511 keV \approx 1.2 keV$

Criteria:

$$\Delta E \ll E_{local,max} \Rightarrow \text{atomic transition (no radiation)},$$

$$\Delta E \lesssim E_{local,max} \Rightarrow \text{outer-layer flux-tube decoupling (radiation)}.$$

This explains why:

- eV-scale transitions necessarily accompany photons;
- yet the electron does not disintegrate.

7.5 Structure and Propagation Speed of Dynamic Magnetism (Hard-Core)

Dynamic Magnetism Is Not "Field Oscillation"

The ontology of dynamic magnetism is: the unfolding in space of the rotational structure of a decoupled magnetic flux tube.

The propagation speed of this structure is determined by the spatial background: $v_{prop} = c$

Not as an approximation, but as a necessary consequence of structural decoupling.

7.6 The Unique Origin of Radiation Frequency (Core Calculation)

7.6.1 Calculation Box 7-1: Ontological Derivation of the Radiation-Frequency Formula

The energy carried by the decoupled dynamic-magnetic structure must equal the transition energy: $E_\gamma = \Delta E$

The energy bookkeeping for dynamic magnetism is: $E_\gamma = \Delta E$

Therefore: $\omega = \frac{\Delta E}{\hbar}$

This is not an assumption; it is energy conservation at the re-enclosure endpoint.

7.6.2 Hydrogen Example (Hard-Core)

Hydrogen ground state to first excited state: $\Delta E = 10.2 V$

Substitute: $\omega = \frac{10.2\times1.602\times10^{-19}}{1.055\times10^{-34}} \approx 1.55 \times 10^{16} s^{-1}$

Corresponding wavelength: $\lambda = \frac{2\pi c}{\omega} \approx 121.6nm$ consistent with the experimental Lyman-α value.

7.7 Gyromagnetic Character of Dynamic Magnetism and the Structural Origin of Polarization

7.7.1 Inheritance of Gyromagnetism

- Static magnetism: rotation is enclosed;
- Dynamic magnetism: rotation is unfolded.

Therefore dynamic magnetism necessarily carries:

- a fixed rotation sense;
- a definite handedness.

7.7.2 Polarization as a Computable Structural Meaning

- Circular polarization: decoupling of a single rotation-sense component;
- Linear polarization: superposition of projected components of two rotation senses;
- Elliptical polarization: unequal amplitudes of rotation-sense components.

Polarization is no longer "the direction of the electric field," but the spatial projection of rotational structure.

7.8 Boundary Between Dynamic Magnetism and Quantization: Continuous Propagation, Discrete Endpoints

Propagation stage:

- dynamic magnetism unfolds continuously;
- it can superpose and interfere.

Re-enclosure endpoint:

- the structure must succeed as a whole;
- "half a decoupled structure" is not allowed.

Thus quantization appears at the re-enclosure endpoint, not during propagation.

7.9 Upper-Bound Theorem of Single-Body Enclosure Mass: Why Proton/Neutron Form the Terminal Level of Stable Particles

From the preceding sections we have established key conclusions: electrons, protons, and neutrons are all single-channel enclosure states organized around a unique principal rotation direction; the proton achieves secondary enclosure on the basis of the electron; the neutron maintains stability at the same level through phase-sign compensation. This section further proves that under the constraints of the enclosure universe (right-handed helical flow and $|\mathbf{g}| = c$), the proton/neutron constitute the upper mass limit for stable "single-body enclosure."

This is not an empirical judgment but a computable and auditable stability criterion.

7.9.1 Co-rotation Mismatch Factor and a Necessary Stability Condition

In Section 3.3.2, the co-rotation relation of the proton relative to the background helical flow was fixed as a dimensionless factor: $\eta_\Omega \equiv \frac{\Omega_{\mathrm{b}g}}{\Omega_p}$

where Ω_{bg} is the background reference rotational frequency at the boundary scale of proton secondary enclosure, and Ω_p is the effective enclosure rotational frequency of the proton shell. A necessary condition for stable single-body enclosure is:

$$\boxed{\eta_\Omega < 1} \tag{7.9.1}$$

Physical meaning: if $\eta_\Omega \geq 1$, the enclosure-shell rotational frequency is not higher than the background reference frequency. Then the screening conditions of right-handed single rotation and minimal 8-period topological closure cannot be maintained; the enclosure disintegrates or can exist only as a non-stable excited state.

7.9.2 The Proton Already Approaches the Co-rotation Limit

In this system, the main-accuracy relation for the proton–electron mass ratio is:

$$\frac{m_p}{m_e} = \frac{\eta_\Omega}{\alpha^2\pi^2} \tag{7.9.2}$$

Substituting the experimental values $m_p/m_e \simeq 1836$and $\alpha^{-1} \simeq 137$, we obtain:

$$\alpha^2\pi^2 \simeq 5.26 \times 10^{-4} \Rightarrow \eta_\Omega \simeq 0.9655 \tag{7.9.3}$$

This means the co-rotation mismatch factor at the proton's secondary-enclosure boundary is already very close to the stability limit $\eta_\Omega = 1$. The neutron introduces a phase-compensation shell at this level and produces only MeV-scale corrections, remaining within the same single-body enclosure level and not exceeding the limit.

7.9.3 Consequence of Assuming a Third Single-Body Enclosure Level

If one assumes that the single-body enclosure hierarchy can continue upward, then without introducing new enclosure mechanisms the most natural recursion is to apply the same main enclosure multiplier again:

$$\frac{m_{(3)}}{m_p} \approx \frac{1}{\alpha^2\pi^2} \tag{7.9.4}$$

Thus the third-level mass scale would be:

$$\frac{m_{(3)}}{m_e} \approx \frac{m_p}{m_e} \cdot \frac{1}{\alpha^2\pi^2} \approx 1836 \times 1900 \approx 3.5 \times 10^6 \tag{7.9.5}$$

Corresponding energy scale:

$$m_{(3)}c^2 \sim 1.8\, TeV \tag{7.9.6}$$

This TeV scale is not a fit to any existing particle spectrum; it is an endogenous consequence of continuing the enclosure multiplier at the single-body level.

7.9.4 Amplification of the Co-rotation Mismatch and Stability Rejection

In the enclosure universe, the tangential speed is always c, and the single-body geometric scale satisfies:

$$\ell_{eff} \propto \frac{m}{\mu_g}, \quad \Omega \propto \frac{c}{\ell_{eff}} \Rightarrow \Omega \propto \frac{1}{m} \tag{7.9.7}$$

If the third-level mass is larger than the proton by a factor $Q \equiv \frac{m_{(3)}}{m_p} \approx \frac{1}{\alpha^2\pi^2} \approx 1900$,

then its enclosure rotational frequency decreases by the same factor:

$$\Omega_{(3)} \approx \frac{\Omega_p}{Q} \tag{7.9.8}$$

The co-rotation mismatch becomes:

$$\eta_{\Omega,(3)} \frac{\Omega_{\mathrm{b}g}}{\Omega_{(3)}} \approx Q \frac{\Omega_{\mathrm{b}g}}{\Omega_p} Q\, \eta_\Omega \tag{7.9.9}$$

Numerically:

$$\eta_{\Omega,(3)} \approx 1900 \times 0.9655 \approx 1.8 \times 10^3 \gg 1 \tag{7.9.10}$$

This directly violates the necessary stability condition $\eta_\Omega < 1$ (Eq. 7.9.1).

7.9.5 Upper-Bound Theorem for Single-Body Enclosure Mass

Therefore, we obtain:

Under the constraints $|\mathbf{g}| = c$ and right-handed single-channel enclosure screening, the proton/neutron constitute the upper mass limit for stable single-body enclosure.

Any attempt to continue "a third single-body enclosure level" is rejected by the stability criterion because the co-rotation mismatch factor is amplified dramatically.

Higher-mass states can exist only as excited states, resonance states, or composite/statistical enclosures of multiple single bodies, and cannot become new stable fundamental particles.

7.9.6 Note: Distinguishing the Single-Body Limit from Macroscopic Enclosure

It must be emphasized that the upper bound discussed here applies only to single-body particle-level enclosure, under constraints including:

- right-handed single-rotation stability;
- minimal 8-period topological closure;
- co-rotation matching $\eta_\Omega < 1$;
- single-exit visibility.

Stars, planets, and black holes belong to macroscopic statistical enclosures or multi-body collective enclosures, whose stability arises from different enclosure spectra (e.g., the spherically symmetric 8π enclosure) and does not violate the single-body upper-bound theorem.

Role of this subsection in the full book:

it provides, within this framework, the first geometric–dynamical explanation of why the basic particle spectrum terminates at the proton/neutron level, elevating the empirical fact to a stability theorem, and establishing a clear boundary for distinguishing particle-physics enclosure mechanisms from macroscopic astrophysical enclosure mechanisms.

7.10 Chapter Summary (Bridge from Atoms to Electromagnetism)

Atomic transitions are not electron motion but rearrangements of the global single-channel standing-wave structure. When the energy difference cannot be absorbed by local enclosure, the outer-layer structure of the electron flux tube decouples, forming dynamic magnetism that propagates at light speed.

Radiation frequency is uniquely determined by transition energy and Planck's constant; polarization originates from inherited unfolding of rotational structure; and quantization appears at re-enclosure endpoints rather than in propagation.

This chapter uses computable criteria to connect atomic structure seamlessly with the ontology of electromagnetic waves.

Chapter 8 Ontological Structure of Electromagnetic Waves: Helical Space Flow and a Geometric Derivation of Maxwell's Effective Equations

8.1 Goal and Method Statement of This Chapter

This chapter aims to complete the largest logical leap—and the most critical step—of the entire book: to prove that electromagnetism is not an independent fundamental branch of physics, but a set of geometric motion theorems of helical space flow in the propagation state.

To do so, this chapter strictly follows the axiomatic system established in Chapters 1–7 and uses only the following inputs:

1. Ontological axiom: space, as a continuous geometric entity, is intrinsically in motion;
2. Kinematic axiom: the geometric velocity field of space satisfies the unique, non-weakenable magnitude constraint $\boxed{|\mathbf{g}(\mathbf{r},t)| = c}$
3. the three-state structure: enclosure–propagation–re-enclosure;
4. the constraints of single channel and rotational continuity.

The logical chain of this chapter is:

geometric axioms → projection definitions → geometric identities → Maxwell equations → delineation of validity and failure domains

Thereby we complete the repositioning of Maxwell's equations from "physical axioms" to "effective theorems."

8.2 Core Axiom: The Unique Kinematic Constraint of Helical Space Flow

According to the ontological axiom in Chapter 1, space is not a static background but is in continuous motion. This motion is described geometrically by the velocity field $\mathbf{g}(\mathbf{r},t)$, which satisfies the unique and non-weakenable constraint:

$$\boxed{|\mathbf{g}(\mathbf{r},t)| = c}$$

Non-weakenability statement (anti-degeneracy).

This constraint is not an approximation and not a low-energy limit; it is a defining constraint of the ontology of space. Any model that allows $|\mathbf{g}| \neq c$ does not belong to the physical domain discussed in this book. The direct meaning of this axiom is: all variations observed in electromagnetic phenomena are not variations in speed magnitude, but variations in how helical space flow redistributes its direction across three-dimensional space.

8.3 Ontological Repositioning of Field Quantities: Geometric Projections Rather Than Independent Entities

In this system, the "electric field" and "magnetic field" of conventional electromagnetism are no longer regarded as fundamental existents, but are redefined as projection descriptions of the geometric potential of helical space flow.

Definition 8.1 (Magnetic field $\mathbf{B}$) $\boxed{\mathbf{B} \equiv \nabla \times \mathbf{g}}$

Geometric meaning: $\mathbf{B}$ describes the rotational-density distribution of helical space flow, i.e., how rotational structure is distributed in space.

Definition 8.2 (Electric field $\mathbf{E}$) $\boxed{\mathbf{E} \equiv -\partial_t \mathbf{g}}$

Geometric meaning: $\mathbf{E}$ describes the temporal decoupling rate of the rotational structure of helical flow, i.e., the unfolding and rearrangement of rotation along the time dimension.
Non-ontology statement (key).
$\mathbf{E}$ and $\mathbf{B}$ do not have ontological status in this system. They are geometric projection quantities of $\mathbf{g}$, not independently existing "field entities."

8.4 Geometric-Identity Derivation of Maxwell's Equations (Propagation Theorem I)

Theorem 8.1 (Kinematic identity set).
In regions where helical space flow is continuous and has no topological rupture (i.e., the continuous limit of the propagation state), the following relations necessarily hold as geometric identities:
(1) No magnetic monopoles (Gauss's law for magnetism) $\nabla \cdot \mathbf{B} \nabla \cdot (\nabla \times \mathbf{g}) \equiv 0$
Geometric meaning: a continuous helical flow has no head or tail and cannot generate an isolated rotational source.
(2) Faraday's induction law $\nabla \times \mathbf{E} - \partial_t \mathbf{B}$
Geometric meaning: the spatial circulation of the temporal decoupling rate of rotational structure necessarily equals the time variation of rotational density.
These two relations are not dynamical assumptions; they are kinematic theorems directly derived from the projection definitions.

8.5 Enclosure Conservation, Source Terms, and Maxwell's Dynamical Equations (Propagation Theorem II)

In the continuous propagation limit, introduce the enclosure density (defined in Chapter 1):

$$n \equiv \nabla \cdot (\mu_g \, \mathbf{g})$$

Theorem 8.2 (Enclosure conservation and emergence of source terms).
The enclosure density satisfies the continuity equation: $\partial_t n + \nabla \cdot (\mu_g \, \partial_t \mathbf{g}) = 0$
Combining this with $\mathbf{E} = -\partial_t \mathbf{g}$, we obtain:
(1) Gauss's law for electricity $\boxed{\nabla \cdot \mathbf{E} = \frac{n}{\varepsilon_0}}$
Geometric meaning: charge density is not a fundamental postulate, but the divergence manifestation of enclosure endpoints (e.g., electron/proton interfaces) in the continuous limit.
(2) Ampère–Maxwell law $\boxed{\nabla \times B = \mu_0 \mathbf{J} + \mu_0 \varepsilon_0 \, \partial_t \mathbf{E}}$
Geometric meaning: this equation describes how transverse transport of enclosure flow ($\mathbf{J}$) and changes in the decoupling rate jointly sustain the propagation structure of the helical flow.
Constant-relation note: $\varepsilon_0 \mu_0 c^2 = 1$
This is not an empirical input; it is determined by the consistency of $|\mathbf{g}| = c$ with the space-property constant μ_g.

8.6 Continuous–Discrete Endpoint Criterion: From Maxwell Propagation to Re-Enclosure Discrimination

In the preceding sections, Maxwell's equations have been strictly repositioned as geometric projection theorems of helical space flow in the continuous propagation limit. However, to close the logic, a key question remains:

Under what condition does the continuous propagation description fail and transition into discrete re-enclosure discrimination?
This section gives the minimal criterion for this "continuous–discrete switch" without introducing new dynamical assumptions.

Definition 8.1 (Endpoint critical layer)

Assume that within a region, the propagation state of helical space flow can be described by continuous variables (ϕ, Ω, n). If the geometric path length required for phase closure becomes comparable to the structure-relevant length scale, then the region enters the endpoint critical layer. In short: when "the space required for closure" has been exhausted, the continuous description must fail.

Criterion 8.1 (Failure condition of the continuous limit)

A necessary condition for continuous propagation to hold is:
the phase-closure path length ℓ_{close} is significantly larger than the structure-relevant length ℓ_{struct}.
When $\ell_{close} \sim \ell_{struct}$, the continuous limit breaks and the system enters the re-enclosure-dominated regime. This criterion depends only on the relative relation of geometric scales, not on specific materials or parameters.

Proposition 8.2 (Validity domain of Maxwell description)

Maxwell's equations hold only in regions satisfying:

1. ϕ can be treated as a continuously differentiable variable;
2. spatial variation of Ω is slow compared with the structural scale;
3. gradients of n do not exhaust the enclosure-closure budget.

Once any of these conditions is violated, Maxwell's equations are no longer an effective description.

Proposition 8.3 (Re-enclosure threshold mapping)

Within the endpoint critical layer, physical evolution is no longer governed by continuous partial-differential equations but by a threshold-based discrimination mapping:
continuous propagation structure $\longrightarrow$ whether global re-enclosure legality is satisfied $\longrightarrow$ success / failure (discrete outcome).
Mathematically, this mapping is not a differential evolution operator but a global legality-discrimination operator.
Important clarification (to avoid misreading "discontinuous evolution").
The continuous–discrete transition is not a "dynamical jump," but a transition of descriptive level:

- in propagation, quantities are described by continuous fields;
- in the endpoint critical layer, the only physically meaningful question is:
  "Can the global structure still be enclosed successfully?"

Thus the apparent "discontinuity" is not a physical discontinuity, but the loss of applicability of continuous description at that scale.

Corollary 8.3 (Geometric definition of a photon event)

A "photon" is not a fundamental particle in propagation, but a discrete re-enclosure event formed when the propagation state passes the global legality test in the endpoint critical layer. Statistically, this appears as a quantum "click," but its precursor is always a continuous propagation structure.

Role of this subsection.

This subsection closes the logic by:

1. providing a geometric scale criterion for Maxwell-equation failure;
2. showing that no physical contradiction exists between continuous propagation and discrete quantum events;
3. providing a unified mechanism for threshold readout used later in Chapter 10 (entanglement) and Chapter 11 (communication).

Summary (relation to the whole chapter).
By introducing the endpoint critical-layer criterion, Maxwell theory is explicitly restricted to an effective description in the continuous propagation limit, not a cross-scale ultimate axiom. The continuous–discrete boundary becomes unambiguous and assumption-free, thereby completing the logical closure of the electromagnetic-propagation chapter.

8.7 Validity and Failure Domains of Maxwell Theory (Firewall Statement)

Theorem 8.3 (Continuous-limit validity theorem).
Maxwell's equations are valid only when all of the following conditions are satisfied:

1. continuous limit: the helical flow has no explicit topological rupture;
2. non-endpoint region: far from enclosure and re-enclosure endpoints;
3. topological averaging: microscopic "figure-8" topology of the double helix is averaged out;
4. linear regime: rotational perturbations are small amplitude.

Failure-domain warning (strong firewall).
When the scale approaches a single-channel re-enclosure endpoint:

- the partial-derivative description of $\mathbf{E}$ and $\mathbf{B}$ fails;
- Maxwell's equations are no longer applicable;
- the system manifests discrete quantum "click" phenomena.

Decisive statement: continuing to use Maxwell's equations to explain quantum phenomena at the re-enclosure endpoint scale—and thereby producing a "quantum–classical conflict"—does not indicate theoretical incompleteness, but an error of applicability domain.

8.8 Chapter Summary (Repositioning Completed)

This chapter proves that:

1. Maxwell's equations are not bottom-level axioms of the physical world;
2. they are geometric-projection dynamical theorems of helical space flow in the continuous propagation limit;
3. the ontology of electromagnetic waves is a helical propagation structure formed after flux-tube decoupling;
4. charge, current, and field quantities are mathematical expressions of this three-dimensional geometric motion in different dimensions;
5. quantization is not an "exotic patch" of Maxwell theory, but a necessary manifestation of its failure domain.

At this point, electromagnetism has been fully absorbed into the unified framework of helical space flow developed in this book.

Chapter 9 Ontology and Calculation of Quantum Interference: Superposition of Double-Helical Propagation, Endpoint Re-Enclosure, and Statistical Patterns

9.1 Position and Goals of This Chapter

Quantum interference is neither particle splitting nor the ontology of a probabilistic cloud. It is the linear superposition of a three-dimensional double-helical propagation state during the continuous propagation stage; discrete "clicks" occur only at re-enclosure endpoints.

Goals:

- derive interference fringes from the continuous field of double-helical dynamic magnetism;
- explain why stable fringes still form when single particles pass one by one;
- provide auditable calculations of fringe spacing and contrast.

9.2 Minimal Variables and Linear Superposition in the Propagation State

Following Chapter 8, the propagation state is represented by the geometric potential field:

$$\mathbf{g}(\mathbf{r},t) = \mathbf{g}_0(\mathbf{r})\cos\phi\,(\mathbf{r},t) + \mathbf{g}_0^{\perp}(\mathbf{r})\sin\phi\,(\mathbf{r},t)$$

where the phase satisfies the propagation relation:

$$\partial_t\phi + c\,\hat{\mathbf{k}}\cdot\nabla\phi = 0, \quad \omega = \partial_t\phi$$

Key property: during propagation, g obeys linear superposition.

9.3 Double Slit: Double-Helical Superposition of Two Propagation Branches

Let the two paths (two slits) correspond to propagation states $\mathbf{g}_1, \mathbf{g}_2$, with phases ϕ_1, ϕ_2.

The total field is: $\mathbf{g} = \mathbf{g}_1 + \mathbf{g}_2$

Define the observable intensity (statistical weight for endpoint re-enclosure) as:

$$I(\mathbf{r})le|\mathbf{g}|^2\rangle = \langle|\mathbf{g}_1|^2\rangle + \langle|\mathbf{g}_2|^2\rangle + 2\langle\mathbf{g}_1\cdot\mathbf{g}_2\rangle.$$

If the two branch amplitudes are approximately equal (symmetric double slit), then:

$$\boxed{I(\mathbf{r})2A^2[1+\cos\Delta\,\phi(\mathbf{r})]}, \quad \Delta\phi = \phi_1 - \phi_2.$$

The interference term arises from the linear superposition of double-helical propagation states, not from particle self-splitting.

9.4 Fringe Positions and Spacing (Hard-Core Example)

Geometric relation (far field, small angles): $\Delta\phi(\theta) = k\,d\sin\theta\,, \quad k = \frac{\omega}{c} = \frac{2\pi}{\lambda}$.

The maximum condition $\Delta\phi = 2\pi m$ gives: $\boxed{d\sin\theta_m = m\,\lambda}$

For screen distance L and transverse coordinate $y \approx L\theta$: $\boxed{\Delta y = \frac{\lambda L}{d}}$.

Example (applicable to both electrons and light): $\lambda = 500\,nm, L = 1\,m, d = 0.1\,mm \Rightarrow \Delta y = 5\,mm$, consistent with experiment.

9.5 Why Single Particles Passing One by One Still Form Fringes

Propagation stage: $\mathbf{g}$ unfolds continuously and superposes linearly.

Endpoint stage (detector/screen): re-enclosure can only succeed or fail as a whole.

Define the probability density for a single "click" as: $\boxed{P(\mathbf{r})I(\mathbf{r})}$

Repeated independent sampling of $P(\mathbf{r})$ reconstructs $I(\mathbf{r})$ statistically.
Therefore:

- each individual event is discrete;
- the fringe pattern is the statistical manifestation of continuous propagation.

9.6 Stratification of Electron Rotational Frequencies and a Unified Explanation of Atomic Spectra and Radiation

Within this framework, two different hierarchical rotational frequencies in the electron system must be distinguished, to avoid mixing the electron ontology, radiation, and atomic energy levels.
(I) Electron intrinsic rotational frequency (enclosure level)
The electron's double-helical enclosure structure has a unique and locked intrinsic rotational frequency: $\Omega_e = \frac{m_e c^2}{\hbar}$

This frequency determines the electron's existence, mass, and action scale. It belongs to the enclosure level and is non-decouplable and non-radiative. No radiation process involves direct change or dissipation of the electron's intrinsic rotational frequency.
(II) Flux-tube propagation frequency (propagation level)
After decoupling, the outer-layer double helix of the electron flux tube forms a propagation state with frequency: $\omega = \frac{E_\gamma}{\hbar}$, corresponding to the transmissible energy rhythm in a specific radiation process. The two satisfy the simple ratio: $\frac{\omega}{\Omega_e} = \frac{E_\gamma}{m_e c^2}$

Thus all electromagnetic radiation can be understood as fractional rhythms extracted from the electron's enclosure "mother" frequency, rather than disintegration or transition of the electron's ontological structure.
(III) Atomic energy levels and endpoint legality
In atomic systems, discreteness of principal energy levels is not a consequence of propagation laws, but of endpoint phase-closure conditions: only propagation rhythms satisfying closure requirements can be re-enclosed as a radiative event. Fine structure corresponds to second-order differences in endpoint legality among different propagation geometries within the same principal level, and its energy scale is naturally suppressed by α^2. Hyperfine structure arises from residual readout of the nuclear single-channel interface by an (s)-state electron, and its scale is further compressed by α/π and the mass ratio m_e/m_p.
(IV) Key quantitative test: hydrogen ground-state hyperfine structure
For the hydrogen ground state, experiment gives the hyperfine splitting energy:

$$\Delta E_{\text{hfs}}^{\text{exp}} \approx 5.87 \times 10^{-6}\ \text{eV}$$

A theoretical baseline quantity is: $\left(\frac{\alpha}{\pi}\right)\left(\frac{m_e}{m_p}\right) E_1 \approx 1.72 \times 10^{-5}\ \text{eV}$

Their ratio defines the geometric visibility factor of the nuclear interface:

$$C_{\text{vis}} \equiv \quad (\Delta E_h f s^e xp) \quad \left(\frac{\alpha}{\pi}\right)\left(\frac{m_e}{m_p}\right) E_1 \approx 0.34$$

an $O(1)$ dimensionless quantity, indicating that nuclear single-channel interface information, though strongly compressed, still enters the hyperfine spectrum with finite proportion. This also naturally yields the $1/n^3$ scaling for s −states.

(V) Unified conclusion

Therefore, atomic principal levels, fine and hyperfine structures, and electromagnetic radiation can be unified as layered manifestations of the same electron enclosure "mother core," under different decoupling depths and endpoint legality conditions. Radiation is not an ontological electron transition but the global re-enclosure of a propagating flux-tube structure constrained by endpoint legality.

9.7 Chapter Summary: A Unified Physical Picture from Electron Enclosure to Atomic Spectra

This chapter reconstructs atomic structure, radiation processes, and fine/hyperfine spectra from surface description to ontological explanation.

By distinguishing the hierarchy between electron double-helical enclosure and flux-tube propagation, the paper clarifies: the electron ontology does not participate in radiation. The intrinsic enclosure rotational frequency sets the electron's existence and mass scale and is non-decouplable and non-radiative; all electromagnetic radiation arises from decoupling and propagation of the outer layer of the electron flux tube. This removes the long-standing semantic confusion of "electron transitions causing radiation."

Accordingly, atomic principal levels are repositioned as the set of propagation rhythms allowed by endpoint phase closure, rather than the orbital energy of an electron. Spectral discreteness does not arise from quantization of propagation itself, but from endpoint legality filtering. Differences among elemental spectra follow naturally from differences in endpoint enclosure geometry.

Fine and hyperfine structures are unified as graded residuals of the same propagation structure under endpoint legality: fine structure reflects second-order differences in propagation geometry within the visibility layer and is naturally suppressed by α^2; hyperfine structure arises from residual readout of the nuclear single-channel interface by an s −state electron and is further compressed by the nuclear-interface visibility factor and the electron–nucleus mass ratio. The hydrogen 21 cm hyperfine line provides a quantitative test showing that nuclear interface information, though strongly compressed, does not vanish but enters observable spectra with finite geometric proportion.

Thus, a concise unified picture is established: atomic spectra are not trajectories of electron energy transitions, but layered manifestations of the same enclosure–propagation–re-enclosure mechanism under different decoupling depths and endpoint legality conditions. Radiation, energy levels, fine structure, and hyperfine structure become a continuous unfolding of the same geometric mechanism at atomic scale.

This unified interpretation provides a clear physical foundation for later discussions of wholeness, entanglement, and information readout: at all scales, observable outcomes are determined not by propagation itself, but by endpoint legality settlement of the whole structure.

9.8 Why "Which-Path" Information Suppresses Interference

Introducing a path marker is equivalent to coupling the propagation state to external degrees of freedom: $\mathbf{g}_1 \to \mathbf{g}_1 \otimes |M_1\rangle, \quad \mathbf{g}_2 \to \mathbf{g}_2 \otimes |M_2\rangle$

The interference term becomes: $2\langle \mathbf{g}_1 \cdot \mathbf{g}_2 \rangle \langle M_1 | M_2 \rangle$.
If $\langle M_1 | M_2 \rangle = 0$ (fully distinguishable), the interference term vanishes.
Suppression arises because phase correlation is transferred and destroyed, not because of "observer consciousness."

9.9 Interference and Energy Conservation

Propagation energy is continuous.
The minimal unit at the endpoint re-enclosure satisfies: $E = \hbar\omega$.
Hence there is no "energy splitting" problem: energy conservation is settled at re-enclosure endpoints and does not conflict with propagation superposition.

9.10 Correspondence with Traditional Quantum Formulations

- traditional $|\psi|^2 \leftrightarrow I(\mathbf{r})$ (statistical weight of propagation intensity);
- superposition principle ↔ linear superposition of $\mathbf{g}$;
- measurement collapse ↔ endpoint re-enclosure conditions becoming locked.

9.11 Chapter Summary

Quantum interference arises from linear superposition of three-dimensional double-helical propagation states; fringes are determined by phase difference, and their spacing and visibility can be computed directly from geometry and amplitude matching. Discrete clicks from single particles passing one by one are statistical sampling results at re-enclosure endpoints, not particle splitting. Any "which-path" marking suppresses interference by destroying phase correlation.

Chapter 10 A Theorem-Based Derivation of Quantum Entanglement: Non-Decomposability of the Global Double-Helical Shell and the Correlation Function

10.1 Goal of This Chapter and the Logical Pivot

This chapter completes the most critical logical transition of the entire book:
it reduces quantum entanglement from a "supernatural nonlocal phenomenon" to the geometric non-decomposability of a global double-helical shell in the propagation state.
Previously:

- Chapter 8 proved that Maxwell's equations are geometric projection theorems of helical space flow in the continuous limit of propagation;
- and the failure boundary between the continuous limit and re-enclosure endpoints (where quantization occurs) was explicitly delineated.

On this basis, this chapter further proves:

1. an entangled state is not a "correlation between two particles," but a global propagation shell produced by a single physical event;
2. the correlation function $E(\theta) = -\cos\theta$ is not an empirical input, but the unique result of continuous geometric projection under discrete re-enclosure at endpoints;
3. Bell inequalities are not "violated"; rather, their separability premise is geometrically invalid.

10.2 Strict Definition of the Global Double-Helical Shell

Definition 10.1 (Global double-helical shell).

A propagation state produced by a physical interaction (e.g., de-enclosure radiation, pair production, etc.) is called a global double-helical shell if it satisfies the following conditions:

1. Phase single-valuedness: the phase function $\phi(\mathbf{r}, t)$ is continuous and single-valued over the entire propagation shell;
2. Global conservation of rotation sense: during propagation, chirality (rotation direction) exhibits no local breaking;
3. Channel non-decomposability: before endpoint re-enclosure, the propagation state contains no independent channel that can be separated by physical operations.

Necessity statement: if any one of these conditions is violated, the propagation state can be geometrically decomposed into multiple independent substructures, and thus does not constitute an entangled state.

10.3 Non-Decomposability Lemma and the Geometric Failure of Bell Premises

Lemma 10.1 (Uniqueness of the global variable).

At any time, a global double-helical shell can only be described by a single global rotational variable $\widehat{\Omega}$; there exists no physical path that decomposes it into $\widehat{\Omega}_A$ and $\widehat{\Omega}_B$.

Physical-path statement (key).

During the propagation stage satisfying Definition 10.1, there is no local physical operation path that can independently rearrange one endpoint without destroying the global shell.

Corollary 10.1 (Failure of the Bell premise).

The separability premise required by Bell inequalities, $P(A, B|\mathbf{a}, \mathbf{b}, \lambda)P(A|\mathbf{a}, \lambda)\, P(B|\mathbf{b}, \lambda)$ does not hold under the geometry of the global double-helical shell.

This is not a negation of Bell's theorem; it states that the premise of that theorem is geometrically unattainable in this system.

10.4 Geometric Necessity of the Correlation Function $-\cos\theta$ (Core Theorem)

Theorem 10.2 (Continuous projection theorem).

During the propagation stage of a global double-helical shell, any measurement is ontologically equivalent to a geometric projection of the global rotational variable $\widehat{\Omega}$.

Let the measurement direction at endpoint A be $\mathbf{a}$, and at endpoint B be $\mathbf{b}$. The continuous projection values are: $A(\widehat{\Omega}) = \mathbf{a} \cdot \widehat{\Omega}, \quad B(\widehat{\Omega}) = \mathbf{b} \cdot \widehat{\Omega}$.

Statistical averaging (continuous limit)

Under isotropy, averaging over all initial global orientations yields the spherical integral:

$E(\mathbf{a}, \mathbf{b}) \int (\mathbf{a} \cdot \widehat{\Omega})(\mathbf{b} \cdot \widehat{\Omega}) \frac{d\Omega}{4\pi} - \cos\theta$, where θ is the angle between $\mathbf{a}$ and $\mathbf{b}$.

Endpoint re-enclosure and discretization

Experiments do not observe the continuous projection values, but discrete "clicks" at re-enclosure endpoints. The re-enclosure mapping $\mathcal{T}$: $[-1,1] \to \pm 1$ changes only the discretization of outcomes, not the angular dependence.

Uniqueness statement (closing degrees of freedom)

Given global conservation of rotation sense and isotropy, the spherical integral above yields $-\cos\theta$ as the only possible two-body correlation function form.

10.5 Wholeness Is Not Nonlocal Causation: The Firewall Theorem of Causal Consistency

In the preceding sections, quantum entanglement has been repositioned as the non-decomposability of the global double-helical shell. However, any description involving "global structure," if not carefully distinguished, can easily be misread as a form of nonlocal causation or superluminal influence. This section provides a strict separation theorem between wholeness and causation to eliminate such misinterpretations.

Theorem 10.3 (Distinguishing global correlation from controllable causation)

Under the constraints $|\mathbf{g}| = c$, the single-channel principle, and the enclosure–propagation–re-enclosure three-state structure, the non-decomposability of a global structure allows correlation but does not allow controllable causal influence. That is, wholeness determines the structural form of joint outcomes but does not grant any local operation the ability to control the outcome at the other endpoint.

Proof (negation of controllability).

Consider endpoints A and B of a global double-helical shell in propagation:

1. In the propagation state, the global shell has not entered the re-enclosure endpoint; the system has no decidable outcomes;
2. any local operation at one endpoint acts only on the same global variable of the global structure and cannot independently rearrange the outcome space of the other endpoint;
3. when re-enclosure discrimination occurs, the outcome is determined by global legality conditions, not solely by an endpoint's choice.

Therefore, although joint outcomes are structurally correlated, no single-end operation can be used to transmit controllable information.

Corollary 10.3 (No violation of relativistic causality)

Since there exists no controllable information channel:

- the measurement choice at one endpoint does not alter the local statistical distribution at the other endpoint;
- no superluminal signal or causal chain exists;
- relativistic causality remains valid.

The so-called "instant determination" refers only to two consistent readouts of the same global structure, not to physical influence propagation.

Clarification: why "boundary-condition locking" is not nonlocal propagation.

The "boundary condition being determined" should be understood as a global consistency settlement completed by the global structure at re-enclosure. This is ontologically different from "sending a physical influence from A to B":

- the former is a legality discrimination of the global structure;
- the latter requires a controllable, locally tunable transmission mechanism.

This system allows only the former and strictly forbids the latter.

Comparative positioning (not part of the main proof):

Descriptive level	Allowed	Forbidden
Global structure	Correlation	Controllable signal
Local operation	Choice of measurement axis	Control of remote outcome
Causality	Relativistic consistency	Superluminal influence

Corollary 10.3.2 (Correct reading of Bell "violation")
The "violation" of Bell inequalities does not imply nonlocal causation in nature; it implies that the separability premise fails for a globally non-decomposable structure. Hence experimental results do not refute relativistic causality; they refute the assumption that a global structure can be decomposed into independent local random variables.
Role of this subsection.
This subsection:

1. strictly separates "wholeness" from "nonlocal causation" in logic;
2. blocks reviewer attacks that misinterpret this framework as a superluminal theory;
3. provides theoretical protection for the engineering boundary "correlated but uncontrollable" in Chapter 11 (field communication).

Summary (relation to Chapter 10).
With the firewall theorem of causal consistency, entanglement is fully repositioned as geometric correlation of a global structure rather than a superluminal causal mechanism. Chapter 10 is thus closed at the structural, computational, and causal levels.

10.6 Causal Consistency and the Geometric Dissolution of "Nonlocality"
Theorem 10.3 (Causal consistency theorem).
Entanglement correlation is an intrinsic geometric property of the global double-helical shell and does not rely on any superluminal signal transmission. When re-enclosure (measurement) occurs at one endpoint:

- the boundary conditions of the global shell are determined;
- the result at the other endpoint is not "instantaneously influenced," but a necessary manifestation after the global variable has been locked.

Therefore, entanglement does not violate relativistic causality.

10.7 "Downgrading" Entanglement: From a Mystery to a Structural Theorem
In this system:

- entanglement is not an additional assumption;
- not a nonlocal signal;
- not a consciousness effect;

but the non-decomposability of a global double-helical shell produced by a single physical event. The so-called "quantum weirdness" arises from continuing to use separable random variables to describe a global structure in the domain where the continuous limit has already failed.

10.8 Chapter Summary (Structural Closure Completed)
This chapter proves that:

1. quantum entanglement is a geometric necessity of a global propagation shell;
2. the correlation function $-\cos\theta$ arises as the unique result of continuous projection;
3. Bell "violation" originates from geometric failure of separability premises;
4. quantum correlation requires no superluminal mechanism.

At this point, the book completes a full geometric closed loop from enclosure → propagation → re-enclosure → entanglement.

Chapter 11 A Geometric Theoremization of Field Communication: Phase Writing, Propagation, and Re-Enclosure Discrimination

11.1 Goal and Method Statement of This Chapter

In the preceding chapters:

- Chapter 8 repositioned electromagnetism as a geometric theorem of helical space flow in the continuous propagation limit;
- Chapter 10 repositioned quantum entanglement as the non-decomposability of a global double-helical shell.

The goal of this chapter is to complete the first translation from physics to engineering: to prove that "communication" is not a probabilistic collapse event, but a process of writing–propagation–re-enclosure discrimination of phase within a global structure.

Methodologically, this chapter uses only axioms and theorems already established:

- the three-state structure: enclosure–propagation–re-enclosure;
- the single-channel principle and rotational continuity;
- the kinematic axiom $|\mathbf{g}| = c$;
- the non-decomposability of the global shell from Chapter 10.

11.2 Ontological Repositioning of Communication: From "Signal Sending" to "Phase Writing"

Definition 11.1 (Phase writing).

In this system, the minimal operation of communication is not "sending particles" or "modulating amplitudes," but imposing controllable phase boundary conditions on the global structure of the propagation state.

Phase writing means: during endpoint re-enclosure / de-enclosure processes, applying reproducible and distinguishable geometric constraints to the phase ϕ of the global helical structure.

Key point: phase is a structural variable rather than an information label; information is the distinguishability of phase structure.

11.3 Information Carriage in the Propagation State: A Global Variable Rather Than a Local Carrier

Lemma 11.1 (Propagation-state carriage lemma).

During the propagation stage, information is not carried by local particles, but by the phase–rotation configuration of the global double-helical structure.

Proof sketch (geometric).

- the propagation state has no independent endpoints;
- local perturbations can only modify the global structure;
- therefore there exists no physical channel for a "local information packet."

Conclusion: information in the propagation state is a distributed global variable.

11.4 Discrimination and Readout: Threshold Mapping at Re-Enclosure Endpoints

Definition 11.2 (Re-enclosure discrimination).

At an endpoint, a propagation state must re-satisfy enclosure conditions.

This process maps continuous phase projections into discrete outcomes: $\mathcal{T}: \phi \longrightarrow$ enclosable, non-enclosable. This is the only step in communication where a "discrete discrimination" appears.
Theorem 11.1 (Uniqueness of the discrimination rule).
Given fixed endpoint geometry and material conditions, the re-enclosure discrimination rule is unique and admits no multivalued mapping.
Implications:

- "noise" in communication is not an intrinsic property of the propagation state;
- noise originates from instability or uncontrollability of endpoint re-enclosure conditions.

11.5 Feasibility Theorem of Field Communication (Not Superluminal)
Theorem 11.2 (Feasibility of field communication).
If two endpoints share the same global double-helical propagation shell, correlation-based communication can be achieved through phase writing and re-enclosure discrimination.
Important limitations:

- the communication does not carry an independently separable local signal;
- no information transmission bypasses the constraint $|\mathbf{g}| = c$;
- causality is not violated.

Interpretation: endpoints do not "receive a signal," but perform different readouts of the same global structure.

11.6 Why Field Communication Is Not "Superluminal Communication"
Lemma 11.2 (Uncontrollability constraint).
Phase writing can change boundary conditions of the global structure, but cannot select a specific discrete outcome at an endpoint.
Consequences:

- one cannot send arbitrary bits by choosing measurement outcomes;
- only correlation can be established, not a controllable message sequence.

This is fully consistent with the conclusion in Chapter 10 that Bell-type separability premises fail.

11.7 Boundary Between Communication and Computation
Theorem 11.3 (Communication–computation boundary theorem).

- Communication: concerns establishing and reading correlations between endpoints;
- Computation: concerns controllable evolution of phase structure and stable mappings.

Their fundamental difference lies in whether discrete outcomes must be programmably selectable.

11.8 Engineering Feasibility Boundary (Firewall Statement)
Theorem 11.4 (Engineering failure domain).
Field-communication theory fails under the following conditions:

1. endpoints do not share the same global propagation shell;
2. re-enclosure conditions are not repeatable;
3. phase-writing amplitude is insufficient to cross an enclosure threshold;
4. the scale approaches a single space unit (discrete layer dominates).

Decisive statement: continuing to use the concept of "field communication" within these failure domains will lead to irreproducible experimental results. This is not an engineering deficiency but

an error of physical applicability domain.

11.9 Fundamental Differences from Traditional Communication Theory

Traditional communication	Field communication (this framework)
bits / symbols	phase structure
local carriers	global propagation state
noise in the channel	noise at endpoints
copyable	non-copyable
probabilistic error correction	geometric stability

11.10 Chapter Summary (Engineering Translation Completed)

This chapter proves:

1. the ontology of communication is phase writing → propagation → re-enclosure discrimination;
2. information does not propagate as particles or bits, but as global structural variables;
3. field communication establishes correlation, not controllable local messages;
4. causality is not broken throughout the process;
5. engineering feasibility is determined by endpoint geometry and enclosure thresholds.

Thus, this book completes a closed loop from space-enclosure axiom → electromagnetic propagation → entanglement geometry → communication engineering boundaries.

Chapter 12 Geometric Foundations of Continuous Computation: Phase Evolution, Enclosure Thresholds, and Feasibility Boundaries

12.1 Goal of This Chapter and Closing Position of the Book

In the preceding chapters:

- Chapter 8 completed the geometrization of electromagnetism (Maxwell as effective theorems);
- Chapter 10 completed the theoremization of quantum entanglement (non-decomposable wholeness);
- Chapter 11 completed the geometric translation of communication (phase writing–propagation–re-enclosure discrimination).

The goal of this chapter is to complete the final step of the book—and the most disruptive transition: to downgrade "computation" from bit flips and probabilistic rules to a problem of continuous evolution of phase–rotation structure within helical space flow.

This establishes a new computational paradigm:

Computation = Programmable evolution of phase structure under continuous geometric constraints

12.2 Ontological Repositioning of Computation: From Bit States to Geometric States

Definition 12.1 (Geometric state).

In this framework, the "state" of a system is not a discrete bit or a Hilbert-space vector, but a geometric configuration jointly defined by the phase $\phi(\mathbf{r}, t)$ and the rotational structure Ω of

helical space flow. We call this a geometric state.

- A geometric state is continuous;
- its degrees of freedom are constrained by the single-channel principle and rotational continuity;
- arbitrary discrete jumps are not allowed.

12.3 Geometric Definition of a Computational Operation: Phase Evolution

Definition 12.2 (Computational operation).

A single computational operation is equivalent to applying a controllable phase-evolution operator to the geometric state:

$$\phi(\mathbf{r}, t) \longrightarrow \phi(\mathbf{r}, t) + \Delta\phi(\mathbf{r})$$

Key distinction:

- traditional computation: flips symbols (0 ↔ 1);
- continuous computation: smoothly rearranges phase structure.

12.4 Core Lemma of Continuous Computation: Global Controllability

Lemma 12.1 (Global controllability lemma).

Under the conditions of global single-channel structure and rotational continuity, any local phase operation necessarily affects the global geometric state.

Proof sketch (geometric).

- phase is single-valued within the global shell;
- no locally isolatable substructure exists;
- therefore, there is no operation that "acts only on a part without involving the whole."

Implication: continuous computation is naturally globally parallel computation.

12.5 True Origin of Computation and Irreversibility: The Re-Enclosure Threshold

Definition 12.3 (Re-enclosure threshold).

When a geometric state evolves to a critical condition, the system must re-satisfy enclosure conditions and produce a discrete result. This condition is called the re-enclosure threshold.

Theorem 12.1 (Irreversibility theorem).

Irreversibility in computation does not originate from time evolution itself, but from the discretizing mapping at the re-enclosure threshold.

Before the threshold:

- phase evolution is continuous and reversible.

At the threshold:

- a continuous structure is mapped into a discrete outcome;
- irreversibility appears.

This is fully consistent with the discrimination theorem of communication in Chapter 11.

12.6 Why Continuous Computation Naturally Avoids the "Quantum Decoherence" Problem

Theorem 12.2 (Decoherence repositioning theorem).

In traditional quantum computation, "decoherence" arises because discrete-bit descriptions are forcibly applied during the continuous propagation stage.

Within the continuous-computation framework:

- no discrete discrimination is performed during propagation;
- there is no “state collapse”;
- what is called decoherence appears only at re-enclosure endpoints.

Conclusion: continuous computation is naturally immune to decoherence during propagation.

12.7 Geometric Origin of Computational Complexity

Theorem 12.3 (Geometric complexity theorem).

Computational complexity is no longer measured by “gate count,” but is jointly determined by:

1. the topological complexity of phase structure;
2. the number of allowed rotational rearrangements;
3. the number and distribution of enclosure thresholds.

Corollary 12.1 (Naturally efficient problems).

In this framework, the following problems are naturally efficient:

- global-constraint satisfaction problems;
- geometric matching and path-consistency problems;
- large-scale parallel correlation computation.

Corollary 12.2 (Problems that cannot be physically efficient).

The following problems cannot be efficiently realized in physical terms:

- problems requiring frequent discrete discrimination;
- problems that essentially require many locally controllable branching operations.

This yields a physically meaningful boundary for complexity.

12.8 Comparison Between Continuous Computation and Traditional Computation Paradigms

Traditional computation	Continuous computation (this framework)
bit flipping	phase rearrangement
gate operations	geometric evolution
discrete states	continuous geometric states
decoherence problem	discretization only at thresholds
local parallelism	global parallelism

12.9 Engineering Feasibility and Limitations (Firewall)

Theorem 12.4 (Engineering failure domain).

Continuous computation fails under the following conditions:

1. phase writing cannot be controlled with sufficient precision;
2. the global shell is topologically destroyed;
3. the scale approaches a single space unit;
4. re-enclosure thresholds are not repeatable.

Decisive statement: continuing to pursue “continuous-computation advantages” within these failure domains leads to irreproducible and non-verifiable outcomes. This is not a technical issue but an error of physical applicability domain.

12.10 Chapter Summary (Closure Completed)

This chapter proves:

1. the ontology of computation is geometric evolution of phase structure;
2. the continuous evolution stage is naturally parallel, reversible, and decoherence-free;
3. irreversibility originates only from re-enclosure thresholds;
4. computational complexity has explicit physical boundaries.

At this point, Natural Origin IV completes a full closed loop:
space-enclosure axiom → electromagnetic propagation → entanglement geometry → communication engineering → computational paradigm.

Chapter 13 A Unified Picture from Spatial Enclosure to the Physical World

Natural Origin IV completes a systematic reconstruction of the foundational structure of physics. Its goal is not to patch existing theories empirically, but—without introducing new free assumptions—to reduce particles, atoms, electromagnetism, quantum correlations, and information processes to geometric manifestations of spatial enclosure structures across different hierarchical levels.

The starting point of this paper is extremely simple yet decisively constraining: space is not a static background, but a right-handed helical flow at the speed of light, and the magnitude of its geometric velocity field is identically equal to the speed of light, $|\mathbf{g}| = c$.

Under this constraint, all observable differences in the physical world no longer arise from changes in speed magnitude, but from different combinations of direction allocation, rotational continuity, and enclosure conditions.

I. From "Objects" to "Enclosure": The Minimal Definition of Physical Existence

In this framework, objects are no longer treated as independent fundamental entities, but are redefined as stable enclosure states formed by helical space flow under specific conditions. Electrons, protons, and neutrons are unified as single-channel enclosure structures organized around a unique principal rotation: the electron is the minimal enclosure state, the proton completes secondary enclosure, and the neutron maintains stability within the same level through a phase-sign compensation mechanism.

Accordingly, mass is no longer a priori "property," but an equivalent bookkeeping quantity for the energy cost required to maintain enclosure; gravitation is no longer an externally imposed interaction, but a fall effect induced by the spatial distribution of enclosure density n. The single-channel principle and the cross-scale enclosure principal-axis theorem together establish a core fact: any structure that exists stably and interacts externally must possess a unique effective principal axis, inherited across scales in rigid, statistical, or engineering-equivalent forms.

II. From Atoms to the Periodic Table: Structural Allowability Rather Than Empirical Rules

At the atomic scale, "electron orbits" are fully repositioned as a problem of counting standing-wave modes on the global single channel. Through four geometric theorems—phase-closure budgeting, the upper bound on angular complexity, the dimensionality of mode space under spherical symmetry, and two-orientation stability under the single channel—the shell-capacity theorem $N(n) = 2n^2$ is rigorously derived, thereby reducing the periodic table to a geometric dimensional invariant rather than empirical induction or postulated quantum numbers.

This result is not a restatement of quantum-mechanical conclusions; it is derived directly from the legality conditions of enclosure geometry without introducing operators, postulates, or statistical interpretations, constituting an ontological-level explanation of atomic structure.

III. Repositioning Electromagnetism: From Axioms to Effective Theorems

At the propagation level, electromagnetism is reinterpreted as a geometric projection theory of helical space flow in the continuous limit. Electric and magnetic fields are no longer treated as ontological field quantities, but as partial-derivative projections of space flow along time and spatial directions; Maxwell's equations are no longer universal axioms, but effective theorems that necessarily hold only within the continuous propagation limit.

By introducing the endpoint critical-layer criterion, the validity and failure domains of continuous descriptions are explicitly delineated: at re-enclosure endpoints, continuous partial derivatives are no longer applicable, and physical processes become governed by global legality discrimination. The so-called "quantum–classical conflict" is thus repositioned as a confusion of applicability domains rather than a physical contradiction.

IV. Radiation and Spectra: Layered Manifestations of Endpoint Legality

For atomic spectra and radiation, the book gives a unified and concise explanation: radiation is not an ontological transition of the electron, but the global re-enclosure of the electron flux-tube propagation structure under endpoint legality constraints. The enclosure rotational frequency of the electron ontology locks its own existence as the energy "mother core," while propagating radiative rhythms are only fractional components extracted from that mother core.

Atomic principal levels arise from the list of propagation rhythms permitted by endpoints; fine and hyperfine structures are not new interactions, but graded residuals of propagation geometry under endpoint legality. Using the hydrogen 21 cm hyperfine line as a quantitative test, the text shows that nuclear single-channel interface information, though strongly compressed, still enters observable spectra with finite proportion.

V. Wholeness and Causality: The Geometric Theoremization of Entanglement

At the level of quantum correlations, entanglement is strictly repositioned as the non-decomposability of a global double-helical shell. The correlation-function form arises uniquely from continuous geometric projection combined with endpoint discretization, not from nonlocal signal transmission. Through the causal-firewall theorem, the framework clearly distinguishes global correlation from controllable causal influence, thereby ensuring that relativistic causality is not violated.

Entanglement is no longer a synonym for "action at a distance," but the natural manifestation of a global structure before it has been decomposed.

VI. Information and Computation: From Bits to Structural Evolution

At the level of information and computation, the book further argues that communication and computation are not fundamental physical objects, but structural processes of phase writing, propagation, and re-enclosure discrimination. The continuous propagation stage is naturally parallel, reversible, and immune to decoherence; irreversibility appears only at endpoint threshold settlement. In this way, computational complexity acquires explicit physical boundaries, avoiding

abstract paradoxes under purely information-theoretic frameworks.

VII. Stability Boundary: Why the Particle Spectrum Terminates at the Proton Level

By means of the co-rotation mismatch factor and the stability criterion for single-body enclosure, the book proves that under the constraints $|\mathbf{g}| = c$ and right-handed single-channel screening, the proton/neutron constitute the upper mass limit for stable single-body enclosure. Any attempt to stack a "heavier single body" is rejected by stability due to co-rotation mismatch and can appear only as excitation, resonance, or composite enclosure. This elevates the termination of the particle spectrum from an empirical fact to a geometric–dynamical theorem.

VIII. Closing Remark: Physical Unification Comes from Structure, Not from Addition

In summary, Natural Origin IV does not establish a new "list of interactions," but a unified structural picture from spatial enclosure to the physical world. In this picture, matter, fields, quantum correlations, and information processing are no longer independent domains, but natural manifestations of the same enclosure–propagation–re-enclosure mechanism across different scales.

This paper does not claim to have answered all questions, but it provides a clear and auditable path: when we shift foundational questions from "what force acts" to "what structures are allowed to exist," long-separated phenomena begin to exhibit a unified geometric order.

Summary

Under the dual ontology of "objects–space," this work establishes a foundational physical theory system with enclosure–propagation–re-enclosure as the core structure, providing unified and computable ontological explanations for electrons, atoms, atomic nuclei, elemental periodicity, electromagnetic waves, and quantum entanglement. By explicitly locking the computational template and bookkeeping rules in Chapter 1, all subsequent physical objects and phenomena are strictly constrained within the same enclosure-structure framework, thereby avoiding additional ontologies, independent interactions, or empirical assumptions.

In this system, mass is redefined as the minimum energy cost required to maintain enclosure. The electron is proven to be the minimal stable enclosure solution satisfying phase closure and single-channel conditions, and therefore becomes the smallest magnetic dipole in nature. Planck's constant $\hbar$ is anchored as the action scale of the electron's minimal enclosure structure, rather than an independent natural constant. By the single-channel theorem, stable enclosure states are further shown to allow only one external interface, whose spatial manifestation is a closed magnetic flux tube.

At the nuclear level, the proton is strictly established as the second stable level of existence, and its mass scale is locked by the unique admissible dimensionless combination $\alpha^2\pi^2$. Based on the minimization of the composite main-channel energy functional, the neutron is derived as a necessary condition for stable nuclear existence, and the strong nuclear force is redefined as the existence criterion for whether a multi-nucleon system admits a stable composite main channel, rather than a fundamental interaction. The weak interaction is characterized as a reconstruction rule of the compensation shell; the directionality and Q values of β decay follow directly from this rule and agree in scale with experiment.

At the atomic and elemental level, the elemental index Z is given an ontological definition as the number of phase interfaces on the nuclear composite main channel, and atomic "orbits" are reduced to standing-wave modes of the electron flux-tube propagation state on the global single channel. Through strict mode counting, the shell-capacity theorem $2n^2$ is derived without invoking quantum-mechanical operators or empirical rules, yielding the period sequence $2,8,18,32,\ldots$. Valence is defined as the number of single-channel gaps in the outer standing-wave shell; transition elements and deviations in filling order are attributed to second-order energy ordering effects without affecting the capacity theorem.

At the electromagnetic and quantum level, starting from the continuous limit of light-speed helical space flow, Maxwell's equations are strictly established as effective dynamical theorems of three-dimensional helical propagation structure, with explicit validity and failure domains. Electromagnetic waves are reduced to the helical propagation state formed after flux-tube decoupling, and a photon is the minimal re-enclosure unit of that structure at an endpoint. Quantum interference is interpreted as linear superposition in the propagation state, while entanglement is theoremized as the non-decomposability of a global double-helical shell. By a strict integral projection of continuous rotation-sense variables and separation from endpoint threshold discretization, the entanglement correlation function $-\cos\theta$ is derived without introducing hidden variables, and Bell-type separability premises are shown not to hold under this structure.

Overall, this work unifies electrons, atoms, nuclei, elements, electromagnetic waves, and quantum phenomena as expansions of one enclosure-structure system across hierarchical levels, forming a closed theoretical framework composed of axioms, theorems, and existence criteria. The framework does not rely on additional ontologies or empirical fitting, has a clear logical hierarchy, explicit computational rules, and falsifiability, providing a structured geometric unification perspective for foundational physics and establishing a stable theoretical basis for further study of electricity, information, and continuous computation.

Appendix A Geometric-Projection Derivation of Maxwell's Equations

A.1 Audit Statement and Scope of Applicability

Statement (Audit Statement).

This appendix introduces no new physical assumptions, free parameters, or empirical inputs. All physical premises (helical space flow, the axiom $|\mathbf{g}| = c$, the single-channel principle, rotational continuity, and enclosure conservation) have already been stated in Chapter 8 and in the Introduction.

The sole purpose of this appendix is to formalize the conclusion of Chapter 8—Maxwell's equations as effective theorems arising from geometric projections of helical space flow—into a mathematical derivation that can be audited step by step.

This appendix applies only to:

- the continuous-limit regime of helical space flow;
- regions with no topological rupture and no re-enclosure endpoints;
- the non-discretized level (i.e., not the level of quantum "clicks").

A.2 Mathematical Objects and Notation

Let space be a three-dimensional differentiable manifold $\mathcal{M} \subset \mathbb{R}^3$, and let time be a parameter $t \in \mathbb{R}$.

Define the geometric velocity field: $\mathbf{g}(\mathbf{r}, t): \mathcal{M} \times \mathbb{R} \to \mathbb{R}^3$

satisfying the axiom of the main text: $\boxed{|\mathbf{g}(\mathbf{r}, t)| = c}$

Define the following geometric projection quantities (used only as mathematical variables):

$\mathbf{B} := \nabla \times \mathbf{g}, \quad \mathbf{E} := -\partial_t \mathbf{g}$

Important note.

Here $\mathbf{E}$ and $\mathbf{B}$ are only partial-derivative projection variables of $\mathbf{g}$; they are not assigned any ontological field meaning.

A.3 Geometric Identity for the No-Magnetic-Monopole Law

Proposition A.1. $\nabla \cdot \mathbf{B} = 0$

Proof.

By definition, $\nabla \cdot \mathbf{B} \nabla \cdot (\nabla \times \mathbf{g})$

For any sufficiently smooth vector field $\mathbf{g}$, the vector-calculus identity holds: $\nabla \cdot (\nabla \times \mathbf{g}) \equiv 0$.

Hence the proposition is proved.

Audit remark.

This is a purely mathematical identity and is independent of any physical assumption.

A.4 Formal Derivation of Faraday's Induction Law

Proposition A.2. $\nabla \times \mathbf{E} = -\partial_t \mathbf{B}$ Proof.

By definition, $\nabla \times \mathbf{E} \nabla \times (-\partial_t \mathbf{g}) - \partial_t (\nabla \times \mathbf{g})$,

where we used the commutativity of time differentiation and spatial curl in the continuous limit.

Substituting $\mathbf{B} = \nabla \times \mathbf{g}$ yields $\nabla \times \mathbf{E} = -\partial_t \mathbf{B}$

A.5 Enclosure Conservation and the Charge-Divergence Term

Define the enclosure density (as in Chapter 8): $n := \nabla \cdot (\mu_g \, \mathbf{g})$, where μ_g is a space-property constant.

Proposition A.3 (Enclosure continuity)

The enclosure density satisfies the continuity equation: $\partial_t n + \nabla \cdot (\mu_g \, \partial_t \mathbf{g}) = 0$

Proof.

Differentiate n with respect to time: $\partial_t n \nabla \cdot (\mu_g \, \partial_t \mathbf{g})$,

Writing it in continuity form gives $\partial_t n + \nabla \cdot (\mu_g, \partial_{t\mathbf{g}}) = 0$.

A.6 Geometric Form of Gauss's Law for Electricity

Since $\mathbf{E} = -\partial_t \mathbf{g}$, Proposition A.3 can be written as $\nabla \cdot \mathbf{E} = \frac{n}{\varepsilon_0}$, where the constant ε_0 is

defined as $\varepsilon_0 := \frac{1}{\mu_g c^2}$

Proposition A.4 (Gauss's law for electricity) $\boxed{\nabla \cdot \mathbf{E} = \frac{n}{\varepsilon_0}}$

Audit remark.

Charge density is not an independent postulate; it is the divergence manifestation of enclosure endpoints in the continuous limit.

A.7 Formal Appearance of the Ampère–Maxwell Law

Define the enclosure-flow density: $\mathbf{J} := \mu_g\, \partial_t \mathbf{g}$.

From $\mathbf{B} = \nabla \times \mathbf{g}$, differentiating with respect to time and combining the above definitions yields: $\nabla \times \mathbf{B}\mu_0\mathbf{J} + \mu_0\varepsilon_0\, \partial_t \mathbf{E}$, where $\mu_0 := \mu_g$.

Proposition A.5 (Ampère–Maxwell law) $\nabla \times \mathbf{B}\mu_0\mathbf{J} + \mu_0\varepsilon_0\, \partial_t \mathbf{E}$

A.8 Constant Relation and Consistency with the Speed of Light

From the definitions $\varepsilon_0 = \frac{1}{\mu_g c^2}$, $\mu_0 = \mu_g$, we obtain directly: $\boxed{\varepsilon_0\mu_0 c^2 = 1}$

Audit remark.

This relation is not an experimental input; it is fixed by the consistency between the axiom $|\mathbf{g}| = c$ and the definition of enclosure density.

A.9 Validity Statement (Failure-Domain Boundary)

All derivations in this appendix assume:

- $\mathbf{g}$ is continuously differentiable;
- there are no enclosure/re-enclosure endpoints;
- there is no topological rupture or discretization.

When the system scale approaches a single-channel re-enclosure endpoint:

- the partial-derivative definitions fail;
- the continuous descriptions of $\mathbf{E}$ and $\mathbf{B}$ no longer hold;
- Maxwell's equations are not applicable.

A.10 Appendix Conclusion

Through strict vector analysis and continuity derivations, this appendix proves that:

- Maxwell's equations can be derived entirely from geometric projections of the helical space-flow field $\mathbf{g}$ together with enclosure conservation;
- $\mathbf{E}$ and $\mathbf{B}$ are not ontological field quantities, but effective variables in the continuous limit of propagation;
- Maxwell theory necessarily fails in the discrete re-enclosure regime.

This result forms a "physical interpretation—mathematical audit" correspondence with Chapter 8.

Appendix B Formal Geometric Proof of the Shell-Capacity Theorem $2n^2$

B.1 Audit Statement and Scope of Applicability

Statement (Audit Statement).

This appendix introduces no new physical assumptions, free parameters, or empirical inputs. All physical premises (single channel, principal rotation, phase closure, rotational continuity) and symbol definitions have already been given in Chapter 6 and in the Introduction.

The sole purpose of this appendix is to formalize the key proposition of Chapter 6—the shell-capacity theorem $N(n) = 2n^2$—into an auditable geometric–topological proof.

Scope.
This appendix applies only to:

- global single-channel enclosure structures;
- spherically symmetric boundary conditions;
- stable standing waves (not propagation states and not endpoint re-enclosure states).

B.2 Closed Mapping and Topological Definition of Radial Order

Let the global single channel be a closed curve Γ, and let the phase function $\phi: \Gamma \to S^1$ be continuous and single-valued.
Definition B.1 (Winding number).
The phase-closure condition along Γ is written as $\oint_\Gamma \nabla \phi \cdot d\ell = 2\pi n, \quad n \in \mathbb{Z}^+$.
Here n is the winding number of the mapping $\phi: \Gamma \to S^1$.
Property B.1.

- n is a topological invariant;
- under continuous deformation that preserves phase single-valuedness, n is unchanged;
- n specifies the available phase-closure budget on the global single channel.

B.3 Angular Complexity and the Upper Bound of Closure Budget

Consider standing-wave solutions under spherical symmetry. Decompose the modes by angular complexity, with angular order $\ell \in \mathbb{Z}_{\geq 0}$.
Lemma B.1 (Angular-complexity cost).
On a global single channel, introducing one independent angular node necessarily introduces a non-removable phase winding.
Reason: an angular node corresponds to a nontrivial phase loop on the sphere; any closed path contributes at least one 2π winding.
Theorem B.1 (Upper bound on angular complexity).
For a global single channel with radial closure order n, the realizable angular complexity satisfies

$$\boxed{\ell \leq n - 1}$$

Proof.

- By Definition B.1, the total phase-closure budget is n;
- by Lemma B.1, each additional angular node consumes one closure unit;
- if $\ell \geq n$, angular consumption is not less than the total budget, so radial closure cannot be completed;
- contradiction with phase single-valuedness.

Hence $\ell \leq n - 1$.

B.4 Dimension Theorem of Spherical-Harmonic Subspaces

Under spherical symmetry, angular standing-wave modes are given by the spherical-harmonic function space.
Theorem B.2 (Dimension of angular mode space).

A spherical harmonic subspace of order ℓ, with dimensions of: $\boxed{\dim \mathcal{H}_\ell = 2\ell + 1}$

Remark: this is a spectral-theory result of the spherical Laplacian and is purely mathematical, independent of any quantum-mechanical interpretation.

B.5 Topological Constraint from Single-Channel Continuity

Let a particular angular mode $\psi_{\ell m}$ be selected.

Definition B.2 (Enclosure orientation).

An enclosure orientation is the choice of helical direction along the principal-rotation axis Ω.

Theorem B.3 (Two-orientation stability).

Under single-channel continuity constraints, any given spatial standing-wave mode allows at most two stable enclosure orientations.

Proof (by contradiction).

1. Assume three or more mutually independent enclosure orientations exist;
2. then at least three inequivalent rotational loops exist;
3. the single-channel constraint requires these loops to merge into one continuous channel;
4. at the merging point, phase multivaluedness or rotational discontinuity must occur;
5. contradiction with continuity required for stable enclosure.

Thus the third orientation is unstable and can only de-enclose.

Remark (no double counting): $\dim \mathcal{H}_\ell = 2\ell + 1$ counts the azimuthal degeneracy of spatial shape; "two orientations" refer to the two stable helical directions for the same shape along the principal rotation. They belong to different layers of freedom and are not redundant.

B.6 Formal Derivation of the Shell-Capacity Theorem

Theorem B.4 (Shell-capacity theorem).

For a global single-channel enclosure with radial order n, the maximum capacity of stable standing-wave modes is $\boxed{N(n) = 2n^2}$

Proof.

- By Theorem B.1: $\ell = 0,1,\dots,n-1$;
- by Theorem B.2: each ℓ contributes $2\ell + 1$ spatial modes;
- by Theorem B.3: each spatial mode allows 2 stable orientations.

Therefore:

$$N(n)\& = 2\sum_{\ell=0}^{n-1}(2\ell + 1)\& = 2(1 + 3 + 5 + \cdots + (2n - 1)\& = 2n^2$$

B.7 Interpretation as a Geometric Dimensional Invariant

From the above derivation, $2n^2$ is not an empirical sequence but a geometric dimensional invariant of the spherically symmetric standing-wave mode space under a global single-channel enclosure background. It depends only on:

1. the topological constraint of phase closure;
2. the geometric structure of spherical symmetry;
3. the stability requirement of single-channel continuity;

and does not depend on any quantum-mechanical operators, postulates, or statistical assumptions.

B.8 Appendix Conclusion

This appendix provides a formal geometric–topological proof of $N(n) = 2n^2$. The theorem corresponds one-to-one with the physical interpretation in Chapter 6, but is logically independent and serves as an audit-level mathematical support.

Appendix C Framework Firewall: Concept Clarification, Boundary Conditions, and Exclusion of Common Misreadings

C.1 Audit Statement and Role of This Appendix

Statement (Firewall Statement).

This appendix introduces no new physical assumptions, theorems, or derivational results. Its sole purpose is to clarify the applicability boundaries, concept usage, and explicit negations of the framework established in Natural Origin IV, preventing misinterpretation of this system as a mere "language rewrite" or a "locally equivalent variant" of existing theories.

This appendix should be read together with the main text and Appendices A/B as a protective layer ensuring integrity and auditability.

C.2 This Framework Is Not a "Repackaged Version" of Quantum Mechanics

Negative statement C.1.

The geometric counting, the shell-capacity theorem, and related structures in Natural Origin IV are not a re-expression of the quantum-number system of quantum mechanics. Specifically, this framework neither uses, relies on, nor implicitly contains the following quantum-mechanical postulates or structures:

- no Hilbert space is introduced;
- no state vectors or operators are defined;
- no commutation relations are used;
- no eigenvalue-postulate structure is assumed;
- no probability amplitudes or Born rule are introduced.

The integers n, ℓ used in Chapter 6 and Appendix B are:

- not quantum numbers;
- but geometric bookkeeping quantities of phase-closure winding and angular-complexity budgeting.

Formal similarity does not constitute theoretical equivalence.

C.3 This Framework Does Not Introduce "Fields as Ontology"

Negative statement C.2.

Although the symbols $\mathbf{E}, \mathbf{B}$ appear in the main text and appendices, Natural Origin IV does not admit any independent electromagnetic field ontology.

- $\mathbf{E}, \mathbf{B}$ are only projection variables of the helical space-flow field $\mathbf{g}$;
- Maxwell's equations are strictly repositioned as effective theorems in the continuous propagation limit;
- at re-enclosure endpoints and discrete regimes, descriptions in terms of $\mathbf{E}, \mathbf{B}$ necessarily fail.

Therefore, this framework is not an extension or modification of classical field theory.

C.4 This Framework Does Not Claim Superluminal Causation or Information Transfer

Negative statement C.3.

Although Chapters 10 and 11 discuss entanglement and field communication, Natural Origin IV does not claim any superluminal signal-transmission mechanism.

- non-decomposability of the global double-helical shell is not equivalent to causal influence;
- entanglement correlation arises from geometric constraints of the global structure, not from information exchange;
- field communication establishes correlation readout, not controllable message sending.

All information and energy transmission is strictly constrained by the kinematic axiom $|\mathbf{g}| = c$.

C.5 Clarifying the Concept of "Principal Axis"

Clarification C.1 (The principal axis is not a classical rotation axis).

The "principal axis" in this work is not:

- the rotation axis of a classical rigid body;
- nor a macroscopic angular-momentum vector.

Clarification C.2 (The principal axis is not a superposition of quantum spin vectors).

The principal axis is not:

- an expectation value of a spin operator;
- a simple sum of multi-particle spin vectors.

Correct definition.

The principal axis is the geometric direction that minimizes enclosure cost under enclosure constraints (single channel, rotational continuity, phase closure). It is a minimum-cost direction in the geometric–topological sense, manifested across scales in rigid, statistical, or engineering-equivalent forms.

C.6 Unified Statement on Isotropy and Anisotropy

Clarification C.3.

This framework neither assumes an "essential anisotropy" of nature nor denies isotropy principles.

- without enclosure or with sufficient statistical averaging, the system appears isotropic;
- under stable enclosure, engineering writing, or statistical alignment, the principal axis manifests as an anisotropy direction.

Thus anisotropy is not an added symmetry breaking, but a natural consequence of enclosure structure.

C.7 Applicability and Non-Applicability Domains

Applicability.

The results of Natural Origin IV apply to:

- stable enclosure structures;
- the continuous propagation limit;
- single-channel geometric backgrounds;
- re-enclosure endpoints as the discrete discrimination layer.

Non-applicability.

This framework is not used for:

- strongly random, multi-channel turbulent structures;
- transient systems without a definable principal axis;
- processes approaching a single space unit without statistical averaging.

In such cases, the theorems in the main text should not be extrapolated.

C.8 Statement on Experiments and Engineering

Clarification C.4.

Natural Origin IV does not require "immediately verifiable experimental schemes" as a condition of validity. Its aim is to provide a self-consistent, auditable, and extensible geometric physics framework. The experimental and engineering chapters should be understood as:

- discussions of theoretical allowability;
- not claims of technical maturity.

C.9 Concluding Statement of Appendix C

This appendix explicitly defines the conceptual boundaries, negations, and applicability scope of Natural Origin IV. Its role is not to extend the theory, but to prevent misinterpretation, misuse, or incorrect alignment with existing paradigms. All theorems, corollaries, and engineering discussions in the main text should be understood within the boundaries defined here.

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